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1 Introduction

1.1 A Brief History

In 1958, Jack Kilby built the first integrated circuit flip-flop with two transistors at Texas Instruments. In 2008, Intel's Itanium microprocessor contained more than 2 billion transistors and a 16 Gb Flash memory contained more than 4 billion transistors. This corresponds to a compound annual growth rate of 53% over 50 years. No other technology in history has sustained such a high growth rate lasting for so long.

This incredible growth has come from steady miniaturization of transistors and improvements in manufacturing processes. Most other fields of engineering involve tradeoffs between performance, power, and price. However, as transistors become smaller, they also become faster, dissipate less power, and are cheaper to manufacture. This synergy has not only revolutionized electronics, but also society at large.

The processing performance once dedicated to secret government supercomputers is now available in disposable cellular telephones. The memory once needed for an entire company's accounting system is now carried by a teenager in her iPod. Improvements in integrated circuits have enabled space exploration, made automobiles safer and more fuel-efficient, revolutionized the nature of warfare, brought much of mankind's knowledge to our Web browsers, and made the world a flatter place.

Figure 1.1 shows annual sales in the worldwide semiconductor market. Integrated circuits became a $100 billion/year business in 1994. In 2007, the industry manufactured approximately 6 quintillion (6×10^{18}) transistors, or nearly a billion for every human being on the planet. Thousands of engineers have made their fortunes in the field. New fortunes lie ahead for those with innovative ideas and the talent to bring those ideas to reality.

During the first half of the twentieth century, electronic circuits used large, expensive, power-hungry, and unreliable vacuum tubes. In 1947, John Bardeen and Walter Brattain built the first functioning point contact transistor at Bell Laboratories, shown in Figure 1.2(a) [Riordan97]. It was nearly classified as a military secret, but Bell Labs publicly introduced the device the following year.

> *We have called it the Transistor, T-R-A-N-S-I-S-T-O-R, because it is a resistor or semiconductor device which can amplify electrical signals as they are transferred through it from input to output terminals. It is, if you will, the electrical equivalent of a vacuum tube amplifier. But there the similarity ceases. It has no vacuum, no filament, no glass tube. It is composed entirely of cold, solid substances.*

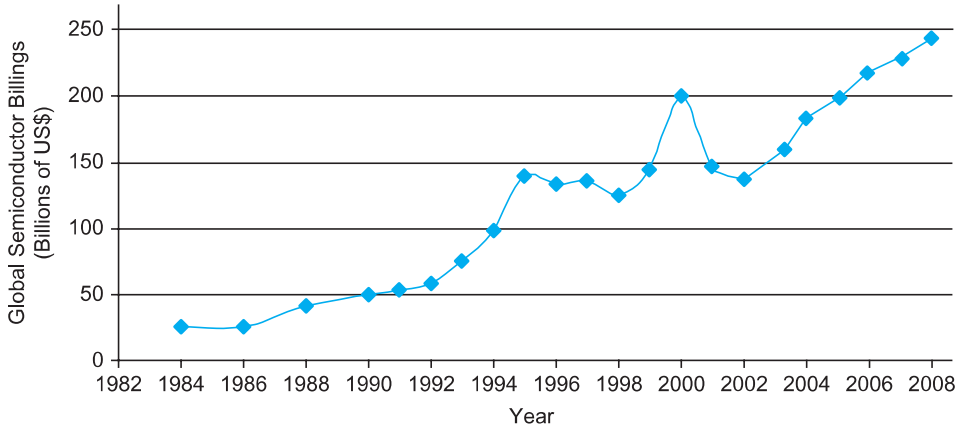
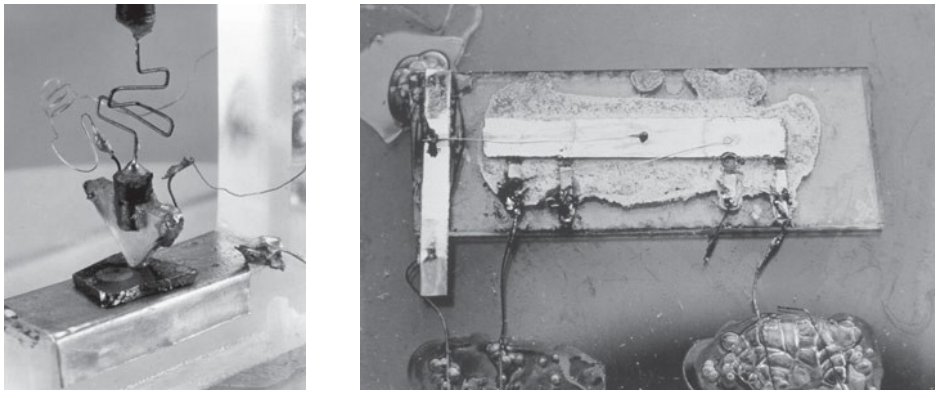


FIGURE 1.1 Size of worldwide semiconductor market (Courtesy of Semiconductor Industry Association.)

Ten years later, Jack Kilby at Texas Instruments realized the potential for miniaturization if multiple transistors could be built on one piece of silicon. Figure 1.2(b) shows his first prototype of an integrated circuit, constructed from a germanium slice and gold wires.

The invention of the transistor earned the Nobel Prize in Physics in 1956 for Bardeen, Brattain, and their supervisor William Shockley. Kilby received the Nobel Prize in Physics in 2000 for the invention of the integrated circuit.

Transistors can be viewed as electrically controlled switches with a control terminal and two other terminals that are connected or disconnected depending on the voltage or current applied to the control. Soon after inventing the point contact transistor, Bell Labs developed the bipolar junction transistor. Bipolar transistors were more reliable, less noisy, and more power-efficient. Early integrated circuits primarily used bipolar transistors. Bipolar transistors require a small current into the control (base) terminal to switch much larger currents between the other two (emitter and collector) terminals. The quiescent power dissipated by these base currents, drawn even when the circuit is not switching,



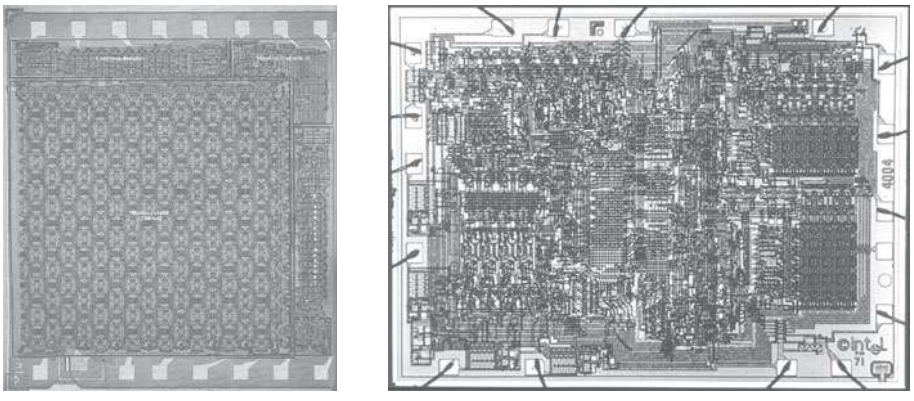
(a) (b)

FIGURE 1.2 (a) First transistor (Property of AT&T Archives. Reprinted with permission of AT&T.) and (b) first integrated circuit (Courtesy of Texas Instruments.)

limits the maximum number of transistors that can be integrated onto a single die. By the 1960s, Metal Oxide Semiconductor Field Effect Transistors (MOSFETs) began to enter production. MOSFETs offer the compelling advantage that they draw almost zero control current while idle. They come in two flavors: nMOS and pMOS, using n-type and p-type silicon, respectively. The original idea of field effect transistors dated back to the German scientist Julius Lilienfield in 1925 [US patent 1,745,175] and a structure closely resembling the MOSFET was proposed in 1935 by Oskar Heil [British patent 439,457], but materials problems foiled early attempts to make functioning devices.

In 1963, Frank Wanlass at Fairchild described the first logic gates using MOSFETs [Wanlass63]. Fairchild's gates used both nMOS and pMOS transistors, earning the name Complementary Metal Oxide Semiconductor, or CMOS. The circuits used discrete transistors but consumed only nanowatts of power, six orders of magnitude less than their bipolar counterparts. With the development of the silicon planar process, MOS integrated circuits became attractive for their low cost because each transistor occupied less area and the fabrication process was simpler [Vadasz69]. Early commercial processes used only pMOS transistors and suffered from poor performance, yield, and reliability. Processes using nMOS transistors became common in the 1970s [Mead80]. Intel pioneered nMOS technology with its 1101 256-bit static random access memory and 4004 4-bit microprocessor, as shown in Figure 1.3. While the nMOS process was less expensive than CMOS, nMOS logic gates still consumed power while idle. Power consumption became a major issue in the 1980s as hundreds of thousands of transistors were integrated onto a single die. CMOS processes were widely adopted and have essentially replaced nMOS and bipolar processes for nearly all digital logic applications.

In 1965, Gordon Moore observed that plotting the number of transistors that can be most economically manufactured on a chip gives a straight line on a semilogarithmic scale [Moore65]. At the time, he found transistor count doubling every 18 months. This observation has been called *Moore's Law* and has become a self-fulfilling prophecy. Figure 1.4 shows that the number of transistors in Intel microprocessors has doubled every 26 months since the invention of the 4004. Moore's Law is driven primarily by *scaling* down the size of transistors and, to a minor extent, by building larger chips. The level of integration of chips has been classified as small-scale, medium-scale, large-scale, and very large-scale. *Small-scale integration* (SSI) circuits, such as the 7404 inverter, have fewer than 10



(a) (b)

FIGURE 1.3 (a) Intel 1101 SRAM (© IEEE 1969 [Vadasz69]) and (b) 4004 microprocessor (Reprinted with permission of Intel Corporation.)

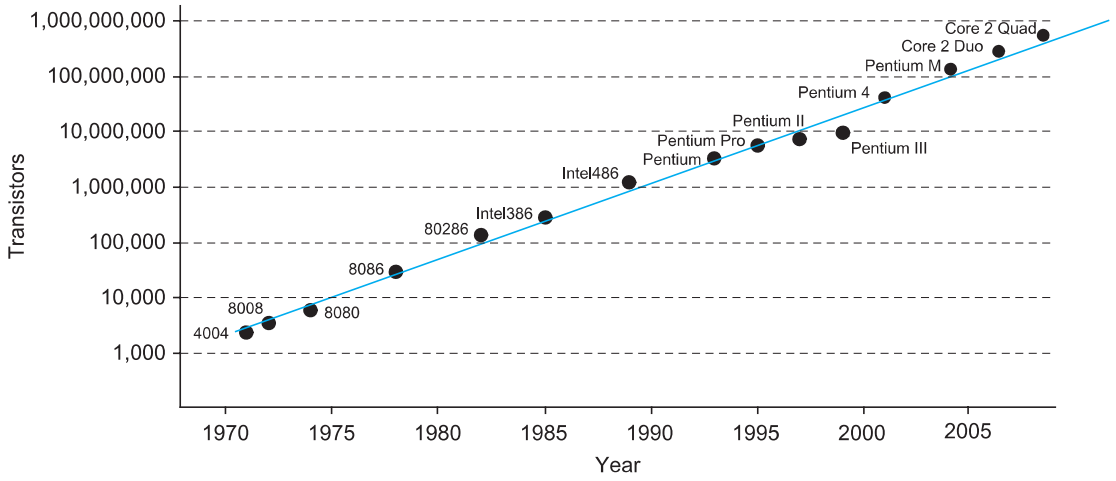


FIGURE 1.4 Transistors in Intel microprocessors [Intel10]

gates, with roughly half a dozen transistors per gate. *Medium-scale integration* (MSI) circuits, such as the 74161 counter, have up to 1000 gates. *Large-scale integration* (LSI) circuits, such as simple 8-bit microprocessors, have up to 10,000 gates. It soon became apparent that new names would have to be created every five years if this naming trend continued and thus the term *very large-scale integration* (VLSI) is used to describe most integrated circuits from the 1980s onward. A corollary of Moore's law is *Dennard's Scaling Law* [Dennard74]: as transistors shrink, they become faster, consume less power, and are cheaper to manufacture. Figure 1.5 shows that Intel microprocessor clock frequencies have doubled roughly every 34 months.This frequency scaling hit the power wall around 2004, and clock frequencies have leveled off around 3 GHz. Computer performance, measured in time to run an application, has advanced even more than raw clock speed. Presently, the performance is driven by the number of cores on a chip rather than by the clock. Even though an individual CMOS transistor uses very little energy each time it switches, the enormous number of transistors switching at very high rates of speed have made power consumption a major design consideration again. Moreover, as transistors have become so small, they cease to turn completely OFF. Small amounts of current leaking through each transistor now lead to significant power consumption when multiplied by millions or billions of transistors on a chip.

The feature size of a CMOS manufacturing process refers to the minimum dimension of a transistor that can be reliably built. The 4004 had a feature size of 10 μm in 1971. The Core 2 Duo had a feature size of 45 nm in 2008. Manufacturers introduce a new process generation (also called a technology node) every 2–3 years with a 30% smaller feature size to pack twice as many transistors in the same area. Figure 1.6 shows the progression of process generations. Feature sizes down to 0.25 μm are generally specified in microns (10^{-6} m), while smaller feature sizes are expressed in nanometers (10^{-9} m). Effects that were relatively minor in micron processes, such as transistor leakage, variations in characteristics of adjacent transistors, and wire resistance, are of great significance in nanometer processes.

Moore's Law has become a self-fulfilling prophecy because each company must keep up with its competitors. Obviously, this scaling cannot go on forever because transistors cannot be smaller than atoms. Dennard scaling has already begun to slow. By the 45 nm

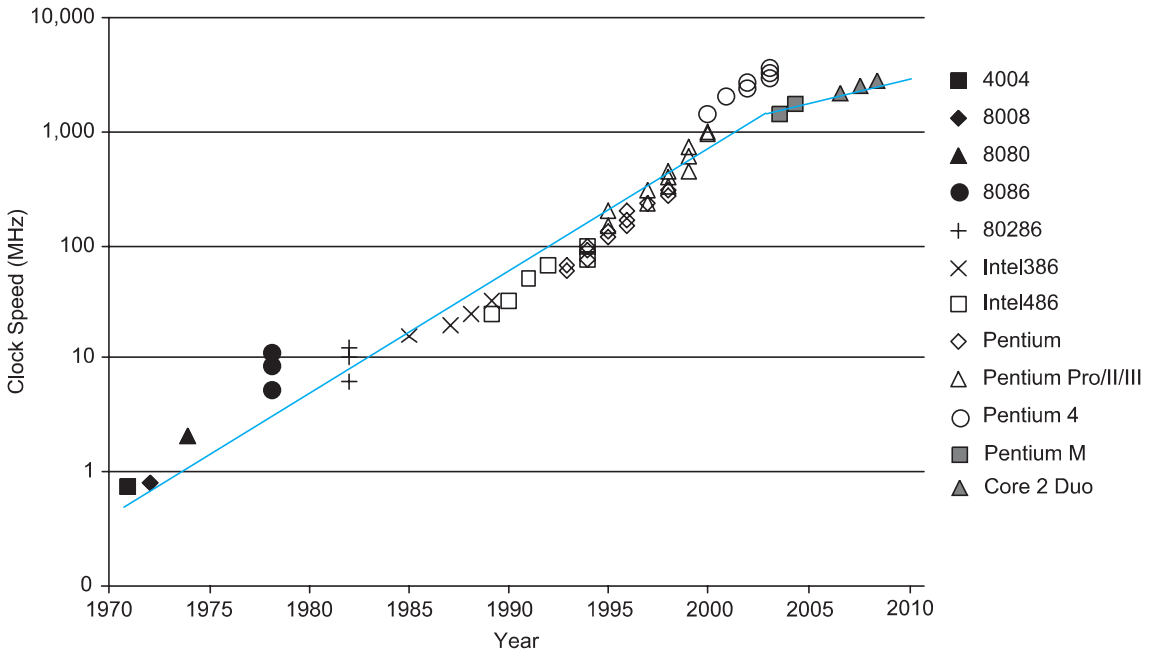


FIGURE 1.5 Clock frequencies of Intel microprocessors

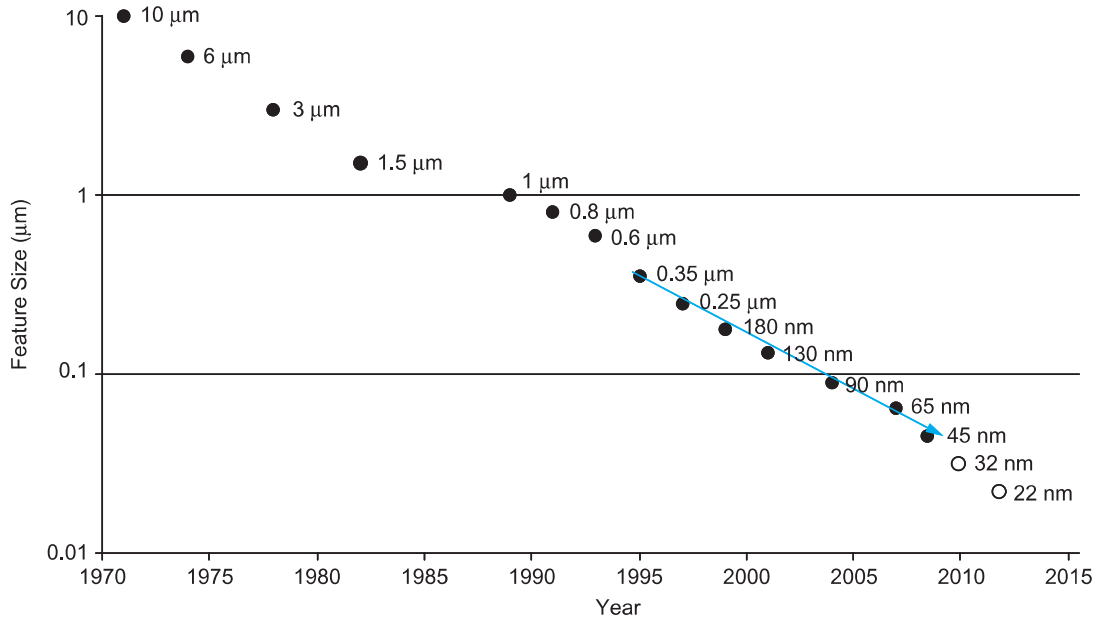


FIGURE 1.6 Process generations. Future predictions from [SIA2007].

generation, designers are having to make trade-offs between improving power and improving delay. Although the cost of printing each transistor goes down, the one-time design costs are increasing exponentially, relegating state-of-the-art processes to chips that will sell in huge quantities or that have cutting-edge performance requirements. However, many predictions of fundamental limits to scaling have already proven wrong. Creative engineers and material scientists have billions of dollars to gain by getting ahead of their competitors. In the early 1990s, experts agreed that scaling would continue for at least a decade but that beyond that point the future was murky. In 2009, we still believe that Moore's Law will continue for at least another decade. The future is yours to invent.

1.2 Preview

As the number of transistors on a chip has grown exponentially, designers have come to rely on increasing levels of automation to seek corresponding productivity gains. Many designers spend much of their effort specifying functions with hardware description languages and seldom look at actual transistors. Nevertheless, chip design is not software engineering. Addressing the harder problems requires a fundamental understanding of circuit and physical design. Therefore, this book focuses on building an understanding of integrated circuits from the bottom up.

In this chapter, we will take a simplified view of CMOS transistors as switches. With this model we will develop CMOS logic gates and latches. CMOS transistors are mass-produced on silicon wafers using lithographic steps much like a printing press process. We will explore how to lay out transistors by specifying rectangles indicating where dopants should be diffused, polysilicon should be grown, metal wires should be deposited, and contacts should be etched to connect all the layers. By the middle of this chapter, you will understand all the principles required to design and lay out your own simple CMOS chip. The chapter concludes with an extended example demonstrating the design of a simple 8-bit MIPS microprocessor chip. The processor raises many of the design issues that will be developed in more depth throughout the book. The best way to learn VLSI design is by doing it. A set of laboratory exercises are available at `www.cmosvlsi.com` to guide you through the design of your own microprocessor chip.

1.3 MOS Transistors

Silicon (Si), a semiconductor, forms the basic starting material for most integrated circuits [Tsividis99]. Pure silicon consists of a three-dimensional *lattice* of atoms. Silicon is a Group IV element, so it forms covalent bonds with four adjacent atoms, as shown in Figure 1.7(a). The lattice is shown in the plane for ease of drawing, but it actually forms a cubic crystal. As all of its valence electrons are involved in chemical bonds, pure silicon is a poor conductor. The conductivity can be raised by introducing small amounts of impurities, called *dopants*, into the silicon lattice. A dopant from Group V of the periodic table, such as arsenic, has five valence electrons. It replaces a silicon atom in the lattice and still bonds to four neighbors, so the fifth valence electron is loosely bound to the arsenic atom, as shown in Figure 1.7(b). Thermal vibration of the lattice at room temperature is enough to set the electron free to move, leaving a positively charged As^+ ion and a free electron. The free electron can carry current so the conductivity is higher. We call this an *n*-type

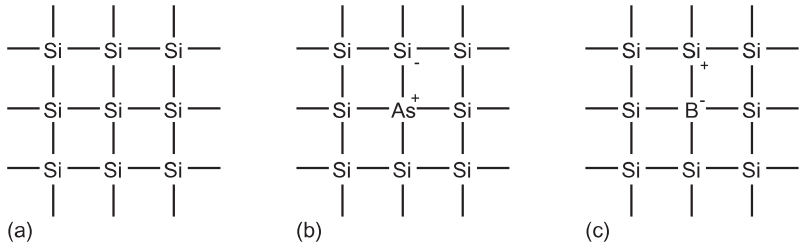


FIGURE 1.7 Silicon lattice and dopant atoms

semiconductor because the free carriers are negatively charged electrons. Similarly, a Group III dopant, such as boron, has three valence electrons, as shown in Figure 1.7(c). The dopant atom can borrow an electron from a neighboring silicon atom, which in turn becomes short by one electron. That atom in turn can borrow an electron, and so forth, so the missing electron, or *hole*, can propagate about the lattice. The hole acts as a positive carrier so we call this a *p*-type semiconductor.

A junction between p-type and n-type silicon is called a *diode*, as shown in Figure 1.8. When the voltage on the p-type semiconductor, called the *anode*, is raised above the n-type *cathode*, the diode is *forward biased* and current flows. When the anode voltage is less than or equal to the cathode voltage, the diode is *reverse biased* and very little current flows.

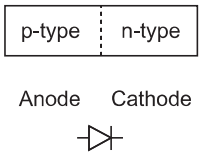


FIGURE 1.8 p-n junction diode structure and symbol

A Metal-Oxide-Semiconductor (*MOS*) structure is created by superimposing several layers of conducting and insulating materials to form a sandwich-like structure. These structures are manufactured using a series of chemical processing steps involving oxidation of the silicon, selective introduction of dopants, and deposition and etching of metal wires and contacts. Transistors are built on nearly flawless single crystals of silicon, which are available as thin flat circular wafers of 15–30 cm in diameter. CMOS technology provides two types of transistors (also called *devices*): an n-type transistor (*nMOS*) and a p-type transistor (*pMOS*). Transistor operation is controlled by electric fields so the devices are also called Metal Oxide Semiconductor Field Effect Transistors (*MOSFETs*) or simply *FETs*. Cross-sections and symbols of these transistors are shown in Figure 1.9. The n+ and p+ regions indicate heavily doped n- or p-type silicon.

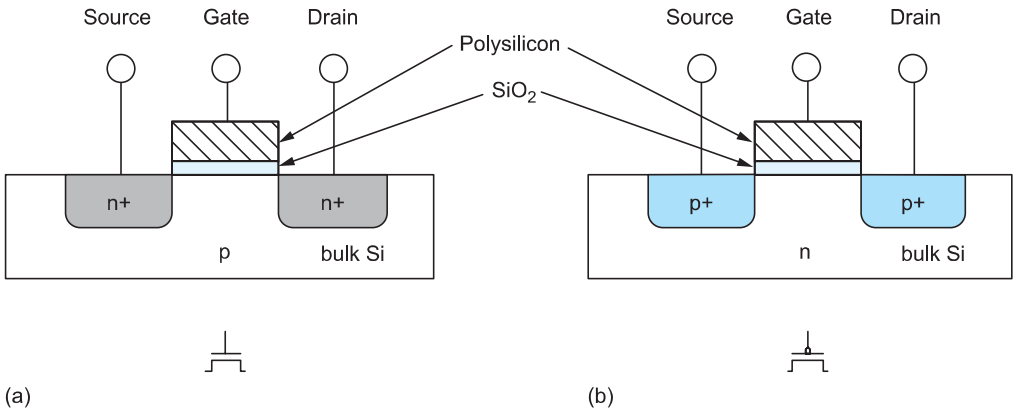


FIGURE 1.9 nMOS transistor (a) and pMOS transistor (b)

Each transistor consists of a stack of the conducting *gate*, an insulating layer of silicon dioxide (SiO_2, better known as glass), and the silicon wafer, also called the *substrate*, *body*, or *bulk*. Gates of early transistors were built from metal, so the stack was called metal-oxide-semiconductor, or MOS. Since the 1970s, the gate has been formed from polycrystalline silicon (*polysilicon*), but the name stuck. (Interestingly, metal gates reemerged in 2007 to solve materials problems in advanced manufacturing processes.) An nMOS transistor is built with a p-type body and has regions of n-type semiconductor adjacent to the gate called the *source* and *drain*. They are physically equivalent and for now we will regard them as interchangeable. The body is typically grounded. A pMOS transistor is just the opposite, consisting of p-type source and drain regions with an n-type body. In a CMOS technology with both flavors of transistors, the substrate is either n-type or p-type. The other flavor of transistor must be built in a special *well* in which dopant atoms have been added to form the body of the opposite type.

The gate is a control input: It affects the flow of electrical current between the source and drain. Consider an nMOS transistor. The body is generally grounded so the p–n junctions of the source and drain to body are reverse-biased. If the gate is also grounded, no current flows through the reverse-biased junctions. Hence, we say the transistor is OFF. If the gate voltage is raised, it creates an electric field that starts to attract free electrons to the underside of the Si–SiO_2 interface. If the voltage is raised enough, the electrons outnumber the holes and a thin region under the gate called the *channel* is inverted to act as an n-type semiconductor. Hence, a conducting path of electron carriers is formed from source to drain and current can flow. We say the transistor is ON.

For a pMOS transistor, the situation is again reversed. The body is held at a positive voltage. When the gate is also at a positive voltage, the source and drain junctions are reverse-biased and no current flows, so the transistor is OFF. When the gate voltage is lowered, positive charges are attracted to the underside of the Si–SiO_2 interface. A sufficiently low gate voltage inverts the channel and a conducting path of positive carriers is formed from source to drain, so the transistor is ON. Notice that the symbol for the pMOS transistor has a bubble on the gate, indicating that the transistor behavior is the opposite of the nMOS.

The positive voltage is usually called V_{DD} or POWER and represents a logic 1 value in digital circuits. In popular logic families of the 1970s and 1980s, V_{DD} was set to 5 volts. Smaller, more recent transistors are unable to withstand such high voltages and have used supplies of 3.3 V, 2.5 V, 1.8 V, 1.5 V, 1.2 V, 1.0 V, and so forth. The low voltage is called GROUND (GND) or V_{SS} and represents a logic 0. It is normally 0 volts.

In summary, the gate of an MOS transistor controls the flow of current between the source and drain. Simplifying this to the extreme allows the MOS transistors to be viewed as simple ON/OFF switches. When the gate of an nMOS transistor is 1, the transistor is ON and there is a conducting path from source to drain. When the gate is low, the nMOS transistor is OFF and almost zero current flows from source to drain. A pMOS transistor is just the opposite, being ON when the gate is low and OFF when the gate is high. This switch model is illustrated in Figure 1.10, where *g*, *s*, and *d* indicate gate, source, and drain. This model will be our most common one when discussing circuit behavior.

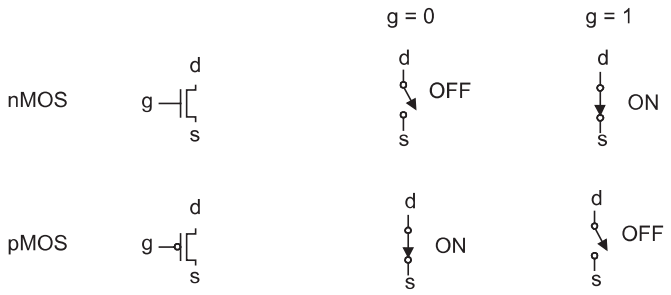


FIGURE 1.10 Transistor symbols and switch-level models

1.4 CMOS Logic

1.4.1 The Inverter

Figure 1.11 shows the schematic and symbol for a CMOS inverter or NOT gate using one nMOS transistor and one pMOS transistor. The bar at the top indicates V_{DD} and the triangle at the bottom indicates GND. When the input A is 0, the nMOS transistor is OFF and the pMOS transistor is ON. Thus, the output Y is pulled up to 1 because it is connected to V_{DD} but not to GND. Conversely, when A is 1, the nMOS is ON, the pMOS is OFF, and Y is pulled down to '0.' This is summarized in Table 1.1.

TABLE 1.1 Inverter truth table

A	Y
0	1
1	0

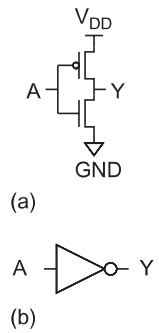


FIGURE 1.11 Inverter schematic (a) and symbol (b) $Y = \overline{A}$

1.4.2 The NAND Gate

Figure 1.12(a) shows a 2-input CMOS NAND gate. It consists of two series nMOS transistors between Y and GND and two parallel pMOS transistors between Y and V_{DD}. If either input A or B is 0, at least one of the nMOS transistors will be OFF, breaking the path from Y to GND. But at least one of the pMOS transistors will be ON, creating a path from Y to V_{DD}. Hence, the output Y will be 1. If both inputs are 1, both of the nMOS transistors will be ON and both of the pMOS transistors will be OFF. Hence, the output will be 0. The truth table is given in Table 1.2 and the symbol is shown in Figure 1.12(b). Note that by DeMorgan's Law, the inversion bubble may be placed on either side of the gate. In the figures in this book, two lines intersecting at a T-junction are connected. Two lines crossing are connected if and only if a dot is shown.

TABLE 1.2 NAND gate truth table

A	B	Pull-Down Network	Pull-Up Network	Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0

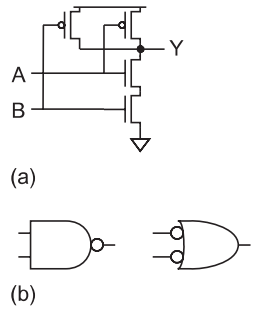


FIGURE 1.12 2-input NAND gate schematic (a) and symbol (b) $Y = \overline{A \cdot B}$

k-input NAND gates are constructed using k series nMOS transistors and k parallel pMOS transistors. For example, a 3-input NAND gate is shown in Figure 1.13. When any of the inputs are 0, the output is pulled high through the parallel pMOS transistors. When all of the inputs are 1, the output is pulled low through the series nMOS transistors.

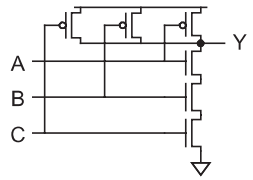


FIGURE 1.13 3-input NAND gate schematic $Y = \overline{A \cdot B \cdot C}$

1.4.3 CMOS Logic Gates

The inverter and NAND gates are examples of *static CMOS logic gates*, also called *complementary CMOS gates*. In general, a static CMOS gate has an nMOS *pull-down network* to connect the output to 0 (GND) and pMOS *pull-up network* to connect the output to 1 (V_{DD}), as shown in Figure 1.14. The networks are arranged such that one is ON and the other OFF for any input pattern.

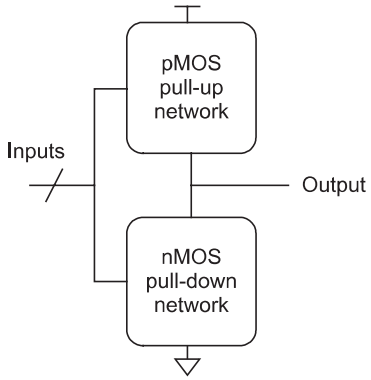


FIGURE 1.14 General logic gate using pull-up and pull-down networks

The pull-up and pull-down networks in the inverter each consist of a single transistor. The NAND gate uses a series pull-down network and a parallel pull-up network. More elaborate networks are used for more complex gates. Two or more transistors in series are ON only if all of the series transistors are ON. Two or more transistors in parallel are ON if any of the parallel transistors are ON. This is illustrated in Figure 1.15 for nMOS and pMOS transistor pairs. By using combinations of these constructions, CMOS combinational gates can be constructed. Although such static CMOS gates are most widely used, Chapter 9 explores alternate ways of building gates with transistors.

In general, when we join a pull-up network to a pull-down network to form a logic gate as shown in Figure 1.14, they both will attempt to exert a logic level at the output. The possible levels at the output are shown in Table 1.3. From this table it can be seen that the output of a CMOS logic gate can be in four states. The 1 and 0 levels have been encountered with the inverter and NAND gates, where either the pull-up or pull-down is OFF and the other structure is ON. When both pull-up and pull-down are OFF, the *high-impedance* or *floating* Z output state results. This is of importance in multiplexers, memory elements, and tristate bus drivers. The *crowbarred* (or *contention*) X level exists when both pull-up and pull-down are simultaneously turned ON. Contention between the two networks results in an indeterminate output level and dissipates static power. It is usually an unwanted condition.

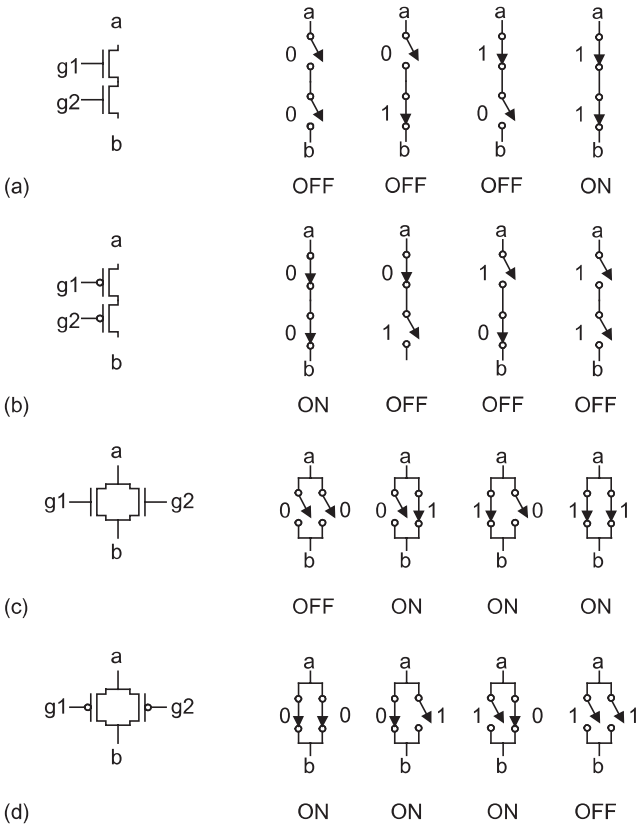


FIGURE 1.15 Connection and behavior of series and parallel transistors

TABLE 1.3 Output states of CMOS logic gates

	pull-up OFF	pull-up ON
pull-down OFF	Z	1
pull-down ON	0	crowbarred (X)

1.4.4 The NOR Gate

A 2-input NOR gate is shown in Figure 1.16. The nMOS transistors are in parallel to pull the output low when either input is high. The pMOS transistors are in series to pull the output high when both inputs are low, as indicated in Table 1.4. The output is never crowbarred or left floating.

TABLE 1.4 NOR gate truth table

A	*B*	*Y*
0	0	1
0	1	0
1	0	0
1	1	0

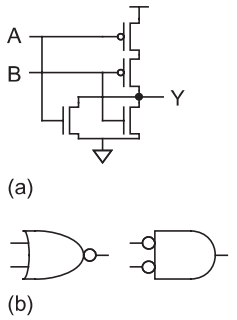


FIGURE 1.16 2-input NOR gate schematic (a) and symbol (b) $Y = \overline{A + B}$

Example 1.1

Sketch a 3-input CMOS NOR gate.

SOLUTION: Figure 1.17 shows such a gate. If any input is high, the output is pulled low through the parallel nMOS transistors. If all inputs are low, the output is pulled high through the series pMOS transistors.

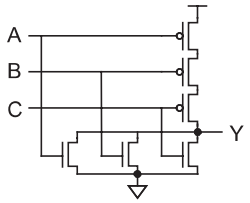


FIGURE 1.17 3-input NOR gate schematic $Y = \overline{A + B + C}$

1.4.5 Compound Gates

A *compound gate* performing a more complex logic function in a single stage of logic is formed by using a combination of series and parallel switch structures. For example, the derivation of the circuit for the function $Y = \overline{(A \cdot B) + (C \cdot D)}$ is shown in Figure 1.18. This function is sometimes called AND-OR-INVERT-22, or AOI22 because it performs the NOR of a pair of 2-input ANDs. For the nMOS pull-down network, take the uninverted expression $((A \cdot B) + (C \cdot D))$ indicating when the output should be pulled to '0.' The AND expressions $(A \cdot B)$ and $(C \cdot D)$ may be implemented by series connections of switches, as shown in Figure 1.18(a). Now ORing the result requires the parallel connection of these two structures, which is shown in Figure 1.18(b). For the pMOS pull-up network, we must compute the complementary expression using switches that turn on with inverted polarity. By DeMorgan's Law, this is equivalent to interchanging AND and OR operations. Hence, transistors that appear in series in the pull-down network must appear in parallel in the pull-up network. Transistors that appear in parallel in the pull-down network must appear in series in the pull-up network. This principle is called *conduction complements* and has already been used in the design of the NAND and NOR gates. In the pull-up network, the parallel combination of A and B is placed in series with the parallel combination of C and D. This progression is evident in Figure 1.18(c) and Figure 1.18(d). Putting the networks together yields the full schematic (Figure 1.18(e)). The symbol is shown in Figure 1.18(f).

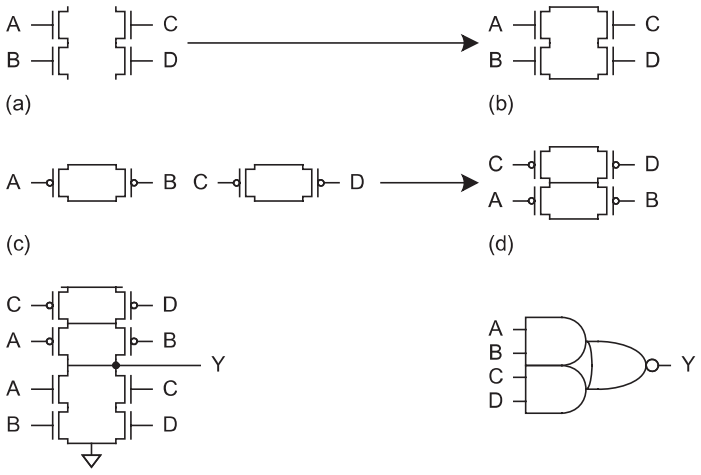


FIGURE 1.18 CMOS compound gate for function $Y = \overline{(A \cdot B) + (C \cdot D)}$

This AOI22 gate can be used as a 2-input inverting multiplexer by connecting $C = \overline{A}$ as a select signal. Then, $Y = \overline{B}$ if C is 0, while $Y = \overline{D}$ if C is 1. Section 1.4.8 shows a way to improve this multiplexer design.

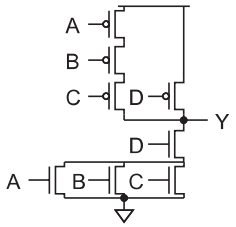


FIGURE 1.19 CMOS compound gate for function $Y = \overline{(A + B + C) \cdot D}$

Example 1.2

Sketch a static CMOS gate computing $Y = \overline{(A + B + C) \cdot D}$.

SOLUTION: Figure 1.19 shows such an OR-AND-INVERT-3-1 (OAI31) gate. The nMOS pull-down network pulls the output low if D is 1 and either A or B or C are 1, so D is in series with the parallel combination of A, B, and C. The pMOS pull-up network is the conduction complement, so D must be in parallel with the series combination of A, B, and C.

1.4.6 Pass Transistors and Transmission Gates

The *strength* of a signal is measured by how closely it approximates an ideal voltage source. In general, the stronger a signal, the more current it can source or sink. The power supplies, or *rails*, (V_{DD} and GND) are the source of the strongest 1s and 0s.

An nMOS transistor is an almost perfect switch when passing a 0 and thus we say it passes a *strong* 0. However, the nMOS transistor is imperfect at passing a 1. The high voltage level is somewhat less than V_{DD}, as will be explained in Section 2.5.4. We say it passes a *degraded* or *weak* 1. A pMOS transistor again has the opposite behavior, passing strong 1s but degraded 0s. The transistor symbols and behaviors are summarized in Figure 1.20 with g, s, and d indicating gate, source, and drain.

When an nMOS or pMOS is used alone as an imperfect switch, we sometimes call it a *pass transistor*. By combining an nMOS and a pMOS transistor in parallel (Figure 1.21(a)), we obtain a switch that turns on when a 1 is applied to g (Figure 1.21(b)) in which 0s and 1s are both passed in an acceptable fashion (Figure 1.21(c)). We term this a *transmission gate* or *pass gate*. In a circuit where only a 0 or a 1 has to be passed, the appropriate transistor (n or p) can be deleted, reverting to a single nMOS or pMOS device.

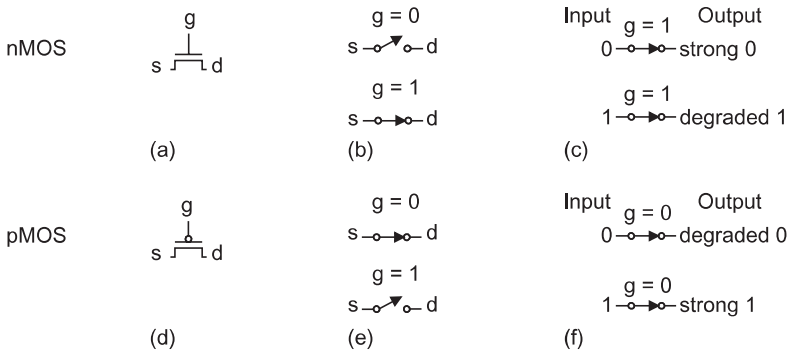


FIGURE 1.20 Pass transistor strong and degraded outputs

Note that both the control input and its complement are required by the transmission gate. This is called *double rail* logic. Some circuit symbols for the transmission gate are shown in Figure 1.21(d).[1] None are easier to draw than the simple schematic, so we will use the schematic version to represent a transmission gate in this book.

In all of our examples so far, the inputs drive the gate terminals of nMOS transistors in the pull-down network and pMOS transistors in the complementary pull-up network, as was shown in Figure 1.14. Thus, the nMOS transistors only need to pass 0s and the pMOS only pass 1s, so the output is always strongly driven and the levels are never degraded. This is called a *fully restored* logic gate and simplifies circuit design considerably. In contrast to other forms of logic, where the pull-up and pull-down switch networks have to be ratioed in some manner, static CMOS gates operate correctly independently of the physical sizes of the transistors. Moreover, there is never a path through 'ON' transistors from the 1 to the 0 supplies for any combination of inputs (in contrast to single-channel MOS, GaAs technologies, or bipolar). As we will find in subsequent chapters, this is the basis for the low static power dissipation in CMOS.

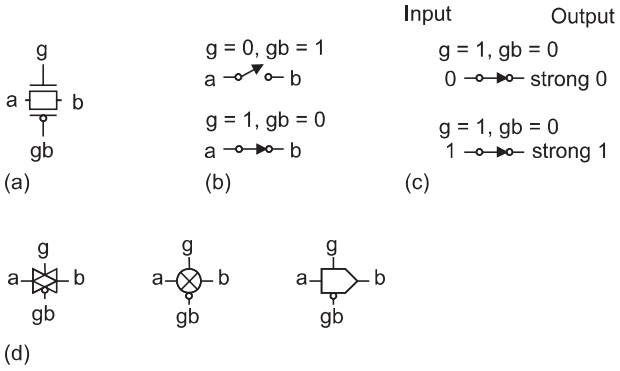


FIGURE 1.21 Transmission gate

[1]We call the left and right terminals *a* and *b* because each is technically the source of one of the transistors and the drain of the other.

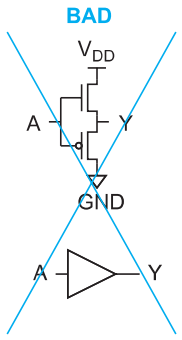


FIGURE 1.22
Bad noninverting buffer

A consequence of the design of static CMOS gates is that they must be inverting. The nMOS pull-down network turns ON when inputs are 1, leading to 0 at the output. We might be tempted to turn the transistors upside down to build a noninverting gate. For example, Figure 1.22 shows a noninverting buffer. Unfortunately, now both the nMOS and pMOS transistors produce degraded outputs, so the technique should be avoided. Instead, we can build noninverting functions from multiple stages of inverting gates. Figure 1.23 shows several ways to build a 4-input AND gate from two levels of inverting static CMOS gates. Each design has different speed, size, and power trade-offs.

Similarly, the compound gate of Figure 1.18 could be built with two AND gates, an OR gate, and an inverter. The AND and OR gates in turn could be constructed from NAND/NOR gates and inverters, as shown in Figure 1.24, using a total of 20 transistors, as compared to eight in Figure 1.18. Good CMOS logic designers exploit the efficiencies of compound gates rather than using large numbers of AND/OR gates.

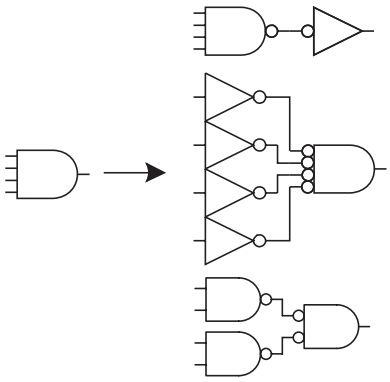


FIGURE 1.23 Various implementations of a CMOS 4-input AND gate

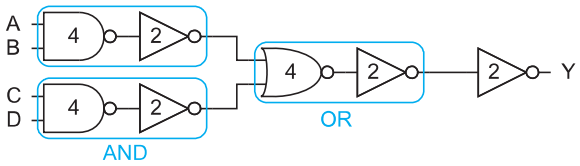


FIGURE 1.24 Inefficient discrete gate implementation of AOI22 with transistor counts indicated

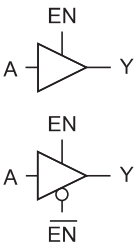


FIGURE 1.25
Tristate buffer symbol

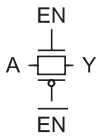


FIGURE 1.26
Transmission gate

1.4.7 Tristates

Figure 1.25 shows symbols for a *tristate buffer*. When the enable input EN is 1, the output Y equals the input A, just as in an ordinary buffer. When the enable is 0, Y is left floating (a 'Z' value). This is summarized in Table 1.5. Sometimes both true and complementary enable signals EN and $\overline{\text{EN}}$ are drawn explicitly, while sometimes only EN is shown.

TABLE 1.5 Truth table for tristate

EN / $\overline{\text{EN}}$	A	Y
0 / 1	0	Z
0 / 1	1	Z
1 / 0	0	0
1 / 0	1	1

The transmission gate in Figure 1.26 has the same truth table as a tristate buffer. It only requires two transistors but it is a *nonrestoring* circuit. If the input is noisy or otherwise degraded, the output will receive the same noise. We will see in Section 4.4.2 that the delay of a series of nonrestoring gates increases rapidly with the number of gates.

Figure 1.27(a) shows a *tristate inverter*. The output is actively driven from V_{DD} or GND, so it is a restoring logic gate. Unlike any of the gates considered so far, the tristate inverter does not obey the conduction complements rule because it allows the output to float under certain input combinations. When EN is 0 (Figure 1.27(b)), both enable transistors are OFF, leaving the output floating. When EN is 1 (Figure 1.27(c)), both enable transistors are ON. They are conceptually removed from the circuit, leaving a simple inverter. Figure 1.27(d) shows symbols for the tristate inverter. The complementary enable signal can be generated internally or can be routed to the cell explicitly. A tristate buffer can be built as an ordinary inverter followed by a tristate inverter.

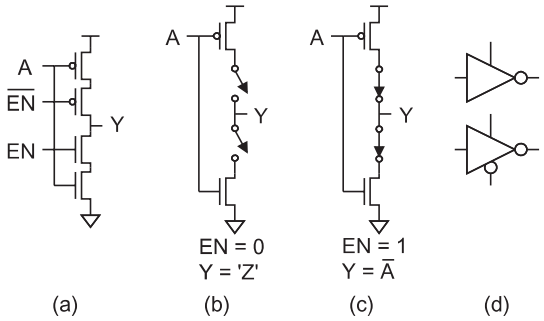


FIGURE 1.27 Tristate Inverter

Tristates were once commonly used to allow multiple units to drive a common bus, as long as exactly one unit is enabled at a time. If multiple units drive the bus, contention occurs and power is wasted. If no units drive the bus, it can float to an invalid logic level that causes the receivers to waste power. Moreover, it can be difficult to switch enable signals at exactly the same time when they are distributed across a large chip. Delay between different enables switching can cause contention. Given these problems, multiplexers are now preferred over tristate busses.

1.4.8 Multiplexers

Multiplexers are key components in CMOS memory elements and data manipulation structures. A *multiplexer* chooses the output from among several inputs based on a select signal. A 2-input, or 2:1 multiplexer, chooses input $D0$ when the select is 0 and input $D1$ when the select is 1. The truth table is given in Table 1.6; the logic function is $Y = \overline{S} \cdot D0 + S \cdot D1$.

TABLE 1.6 Multiplexer truth table

$S / \overline{S}$	$D1$	$D0$	Y
0 / 1	X	0	0
0 / 1	X	1	1
1 / 0	0	X	0
1 / 0	1	X	1

Two transmission gates can be tied together to form a compact 2-input multiplexer, as shown in Figure 1.28(a). The select and its complement enable exactly one of the two transmission gates at any given time. The complementary select $\overline{S}$ is often not drawn in the symbol, as shown in Figure 1.28(b).

Again, the transmission gates produce a nonrestoring multiplexer. We could build a restoring, inverting multiplexer out of gates in several ways. One is the compound gate of Figure 1.18(e), connected as shown in Figure 1.29(a). Another is to gang together two tristate inverters, as shown in Figure 1.29(b). Notice that the schematics of these two approaches are nearly identical, save that the pull-up network has been slightly simplified and permuted in Figure 1.29(b). This is possible because the select and its complement are mutually exclusive. The tristate approach is slightly more compact and faster because it

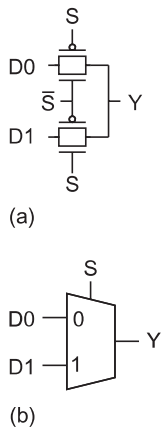


FIGURE 1.28 Transmission gate multiplexer

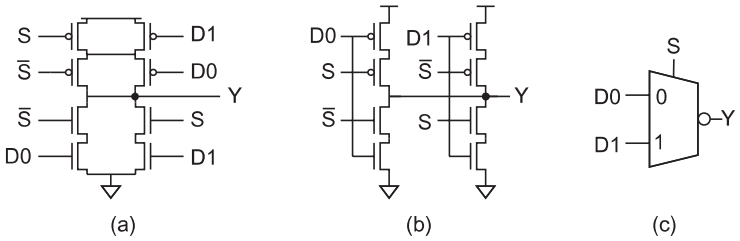


FIGURE 1.29 Inverting multiplexer

requires less internal wire. Again, if the complementary select is generated within the cell, it is omitted from the symbol (Figure 1.29(c)).

Larger multiplexers can be built from multiple 2-input multiplexers or by directly ganging together several tristates. The latter approach requires decoded enable signals for each tristate; the enables should switch simultaneously to prevent contention. 4-input (4:1) multiplexers using each of these approaches are shown in Figure 1.30. In practice, both inverting and noninverting multiplexers are simply called multiplexers or muxes.

1.4.9 Sequential Circuits

So far, we have considered *combinational circuits*, whose outputs depend only on the current inputs. *Sequential circuits* have memory: their outputs depend on both current and previous inputs. Using the combinational circuits developed so far, we can now build sequential circuits such as latches and flip-flops. These elements receive a clock, *CLK*, and a data input, *D*, and produce an output, *Q*. A *D latch* is *transparent* when $CLK = 1$, meaning that *Q* follows *D*. It becomes *opaque* when $CLK = 0$, meaning *Q* retains its previous value and ignores changes in *D*. An *edge-triggered flip-flop* copies *D* to *Q* on the rising edge of *CLK* and remembers its old value at other times.

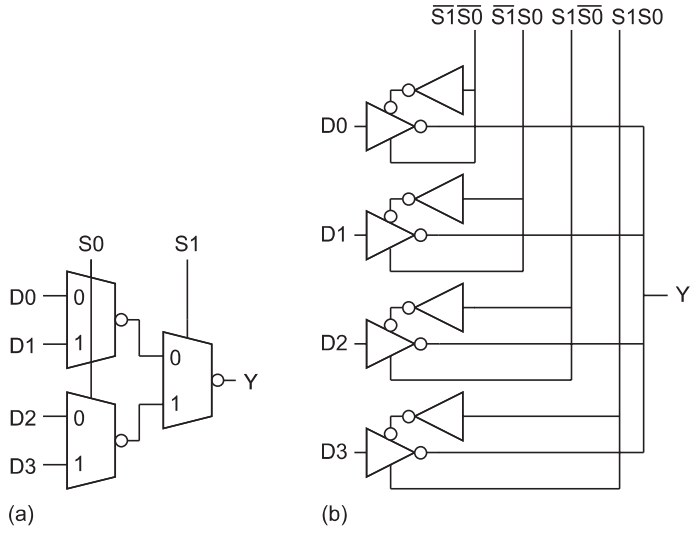


FIGURE 1.30 4:1 multiplexer

1.4.9.1 Latches A *D* latch built from a 2-input multiplexer and two inverters is shown in Figure 1.31(a). The multiplexer can be built from a pair of transmission gates, shown in Figure 1.31(b), because the inverters are restoring. This latch also produces a complementary output, $\overline{Q}$. When $CLK = 1$, the latch is transparent and *D* flows through to *Q* (Figure 1.31(c)). When *CLK* falls to 0, the latch becomes opaque. A feedback path around the inverter pair is established (Figure 1.31(d)) to hold the current state of *Q* indefinitely.

The *D* latch is also known as a *level-sensitive latch* because the state of the output is dependent on the level of the clock signal, as shown in Figure 1.31(e). The latch shown is a positive-level-sensitive latch, represented by the symbol in Figure 1.31(f). By inverting the control connections to the multiplexer, the latch becomes negative-level-sensitive.

1.4.9.2 Flip-Flops By combining two level-sensitive latches, one negative-sensitive and one positive-sensitive, we construct the edge-triggered flip-flop shown in Figure 1.32(a–b). The first latch stage is called the *master* and the second is called the *slave*.

While *CLK* is low, the master negative-level-sensitive latch output ($\overline{QM}$) follows the *D* input while the slave positive-level-sensitive latch holds the previous value (Figure 1.32(c)). When the clock transitions from 0 to 1, the master latch becomes opaque and holds the *D* value at the time of the clock transition. The slave latch becomes transparent, passing the stored master value ($\overline{QM}$) to the output of the slave latch (*Q*). The *D* input is blocked from affecting the output because the master is disconnected from the *D* input (Figure 1.32(d)). When the clock transitions from 1 to 0, the slave latch holds its value and the master starts sampling the input again.

While we have shown a transmission gate multiplexer as the input stage, good design practice would buffer the input and output with inverters, as shown in Figure 1.32(e), to

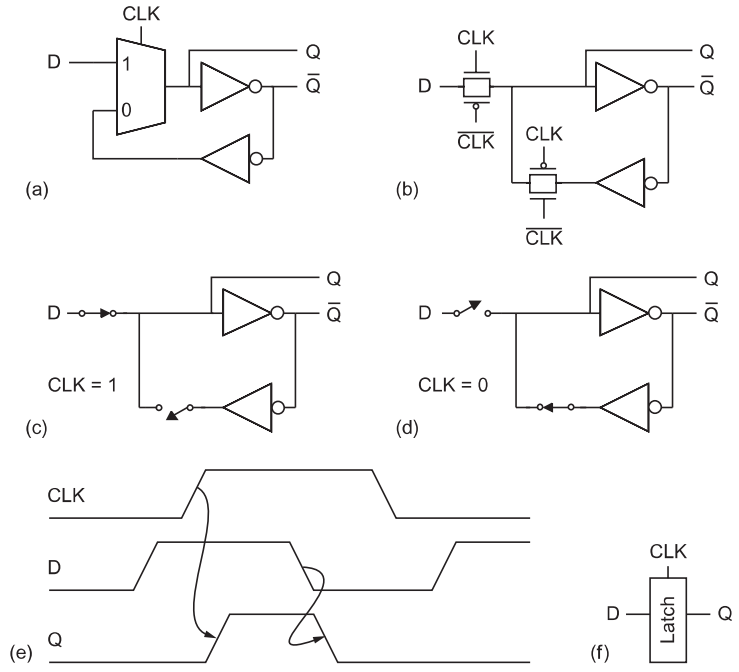


FIGURE 1.31 CMOS positive-level-sensitive *D* latch

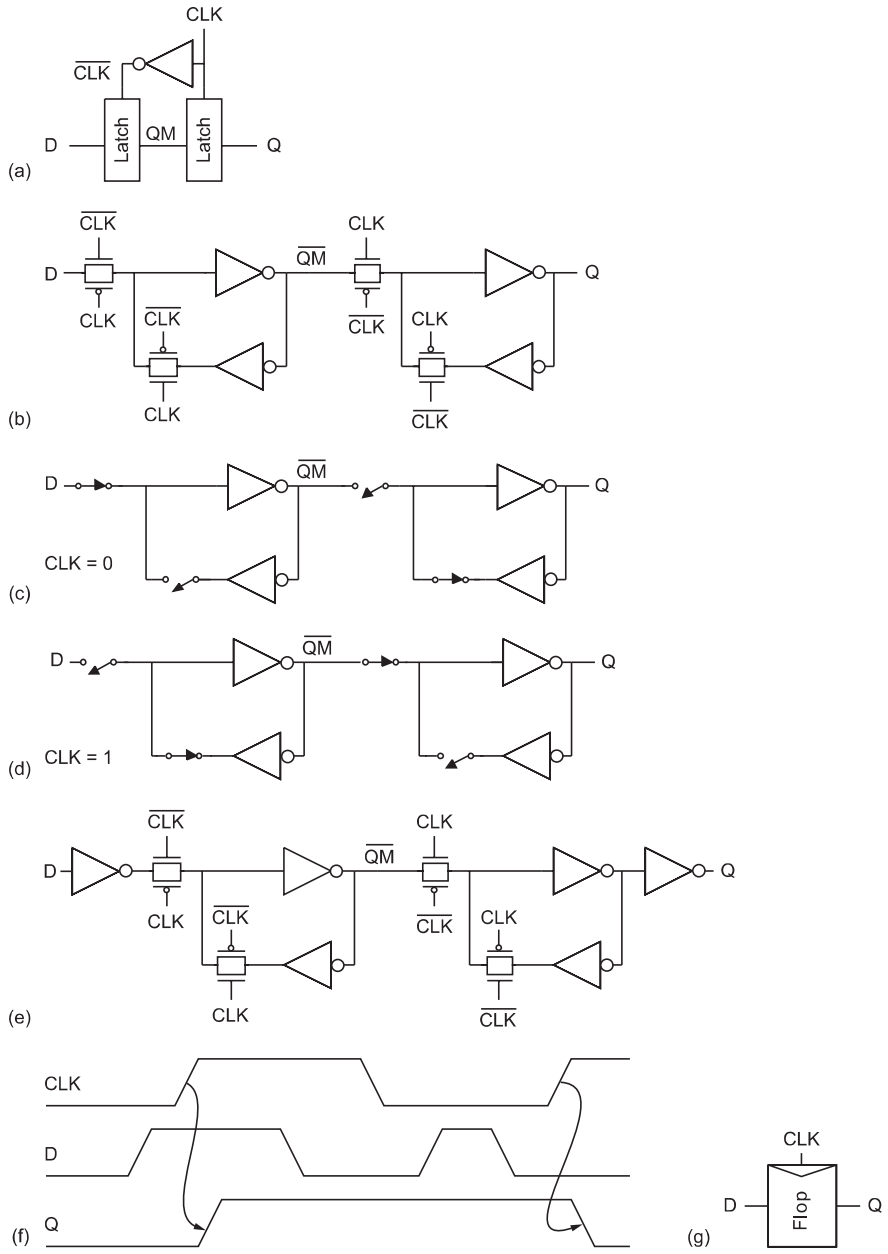


FIGURE 1.32 CMOS positive-edge-triggered *D* flip-flop

preserve what we call "modularity." Modularity is explained further in Section 1.6.2 and robust latches and registers are discussed further in Section 10.3.

In summary, this flip-flop copies *D* to *Q* on the rising edge of the clock, as shown in Figure 1.32(f). Thus, this device is called a positive-edge triggered flip-flop (also called a *D flip-flop*, *D register*, or *master–slave flip-flop*). Figure 1.32(g) shows the circuit symbol for the flip-flop. By reversing the latch polarities, a negative-edge triggered flip-flop may be

constructed. A collection of *D* flip-flops sharing a common clock input is called a *register*. A register is often drawn as a flip-flop with multi-bit *D* and *Q* busses.

In Section 10.2.5 we will see that flip-flops may experience hold-time failures if the system has too much *clock skew*, i.e., if one flip-flop triggers early and another triggers late because of variations in clock arrival times. In industrial designs, a great deal of effort is devoted to timing simulations to catch hold-time problems. When design time is more important (e.g., in class projects), hold-time problems can be avoided altogether by distributing a two-phase nonoverlapping clock. Figure 1.33 shows the flip-flop clocked with two nonoverlapping phases. As long as the phases never overlap, at least one latch will be opaque at any given time and hold-time problems cannot occur.

1.5 CMOS Fabrication and Layout

Now that we can design logic gates and registers from transistors, let us consider how the transistors are built. Designers need to understand the physical implementation of circuits because it has a major impact on performance, power, and cost.

Transistors are fabricated on thin silicon wafers that serve as both a mechanical support and an electrical common point called the *substrate*. We can examine the physical layout of transistors from two perspectives. One is the top view, obtained by looking down on a wafer. The other is the cross-section, obtained by slicing the wafer through the middle of a transistor and looking at it edgewise. We begin by looking at the cross-section of a complete CMOS inverter. We then look at the top view of the same inverter and define a set of masks used to manufacture the different parts of the inverter. The size of the transistors and wires is set by the mask dimensions and is limited by the resolution of the manufacturing process. Continual advancements in this resolution have fueled the exponential growth of the semiconductor industry.

1.5.1 Inverter Cross-Section

Figure 1.34 shows a cross-section and corresponding schematic of an inverter. (See the inside front cover for a color cross-section.) In this diagram, the inverter is built on a p-type substrate. The pMOS transistor requires an n-type body region, so an n-well is diffused into the substrate in its vicinity. As described in Section 1.3, the nMOS transistor

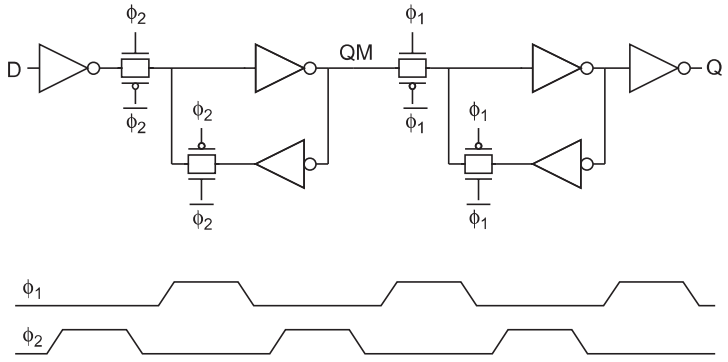


FIGURE 1.33 CMOS flip-flop with two-phase nonoverlapping clocks

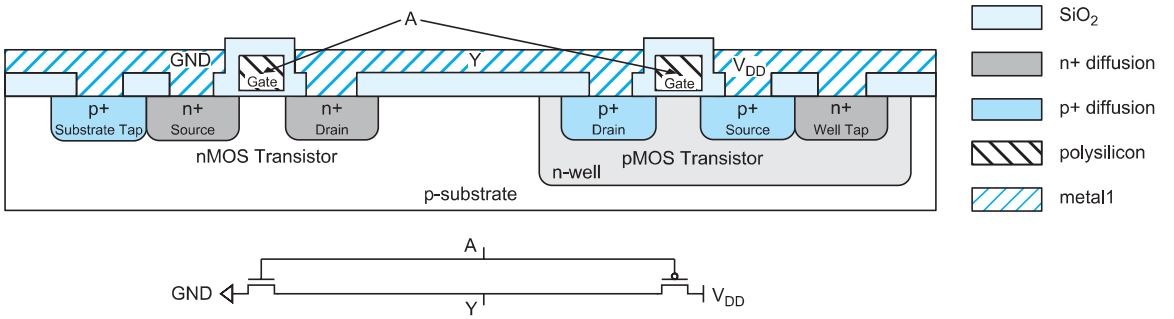


FIGURE 1.34 Inverter cross-section with well and substrate contacts. Color version on inside front cover.

has heavily doped n-type source and drain regions and a polysilicon gate over a thin layer of silicon dioxide (SiO_2, also called *gate oxide*). n+ and p+ diffusion regions indicate heavily doped n-type and p-type silicon. The pMOS transistor is a similar structure with p-type source and drain regions. The polysilicon gates of the two transistors are tied together somewhere off the page and form the input *A*. The source of the nMOS transistor is connected to a metal ground line and the source of the pMOS transistor is connected to a metal V_{DD} line. The drains of the two transistors are connected with metal to form the output *Y*. A thick layer of SiO_2 called *field oxide* prevents metal from shorting to other layers except where contacts are explicitly etched.

A junction between metal and a lightly doped semiconductor forms a *Schottky diode* that only carries current in one direction. When the semiconductor is doped more heavily, it forms a good ohmic contact with metal that provides low resistance for bidirectional current flow. The substrate must be tied to a low potential to avoid forward-biasing the p-n junction between the p-type substrate and the n+ nMOS source or drain. Likewise, the n-well must be tied to a high potential. This is done by adding heavily doped substrate and well contacts, or *taps*, to connect GND and V_{DD} to the substrate and n-well, respectively.

1.5.2 Fabrication Process

For all their complexity, chips are amazingly inexpensive because all the transistors and wires can be printed in much the same way as books. The fabrication sequence consists of a series of steps in which layers of the chip are defined through a process called *photolithography*. Because a whole wafer full of chips is processed in each step, the cost of the chip is proportional to the chip area, rather than the number of transistors. As manufacturing advances allow engineers to build smaller transistors and thus fit more in the same area, each transistor gets cheaper. Smaller transistors are also faster because electrons don't have to travel as far to get from the source to the drain, and they consume less energy because fewer electrons are needed to charge up the gates! This explains the remarkable trend for computers and electronics to become cheaper and more capable with each generation.

The inverter could be defined by a hypothetical set of six masks: n-well, polysilicon, n+ diffusion, p+ diffusion, contacts, and metal (for fabrication reasons discussed in Chapter 3, the actual mask set tends to be more elaborate). Masks specify where the components will be manufactured on the chip. Figure 1.35(a) shows a top view of the six masks. (See also the inside front cover for a color picture.) The cross-section of the inverter from Figure 1.34 was taken along the dashed line. Take some time to convince yourself how the top view and cross-section relate; this is critical to understanding chip layout.

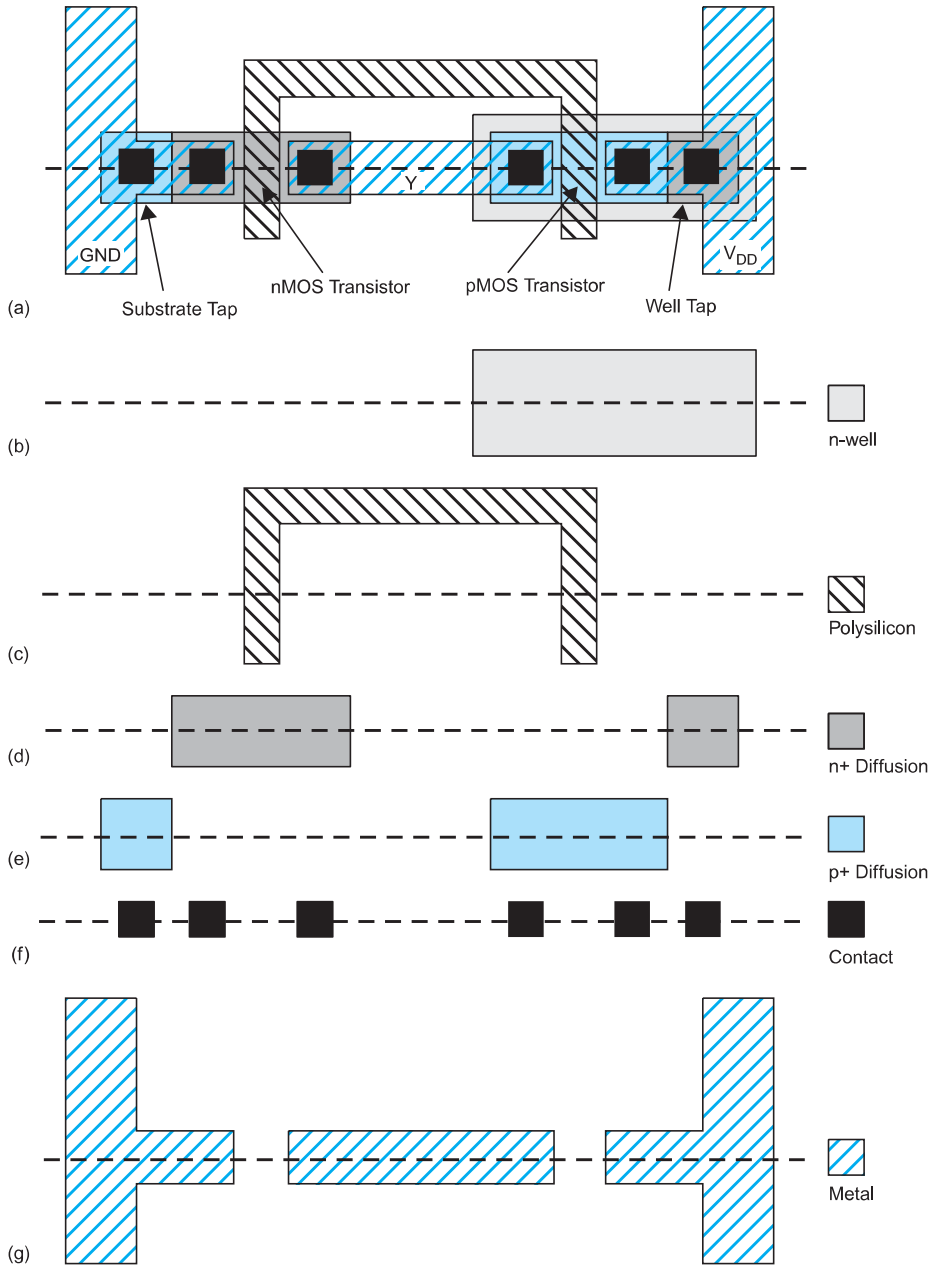


FIGURE 1.35 Inverter mask set. Color version on inside front cover.

Consider a simple fabrication process to illustrate the concept. The process begins with the creation of an n-well on a bare p-type silicon wafer. Figure 1.36 shows cross-sections of the wafer after each processing step involved in forming the n-well; Figure 1.36(a) illustrates the bare substrate before processing. Forming the n-well requires adding enough Group V dopants into the silicon substrate to change the substrate from p-type to n-type in the region of the well. To define what regions receive n-wells, we grow a protective layer of

oxide over the entire wafer, then remove it where we want the wells. We then add the n-type dopants; the dopants are blocked by the oxide, but enter the substrate and form the wells where there is no oxide. The next paragraph describes these steps in detail.

The wafer is first *oxidized* in a high-temperature (typically 900–1200 °C) furnace that causes Si and O_2 to react and become SiO_2 on the wafer surface (Figure 1.36(b)). The oxide must be *patterned* to define the n-well. An organic photoresist[2] that softens where

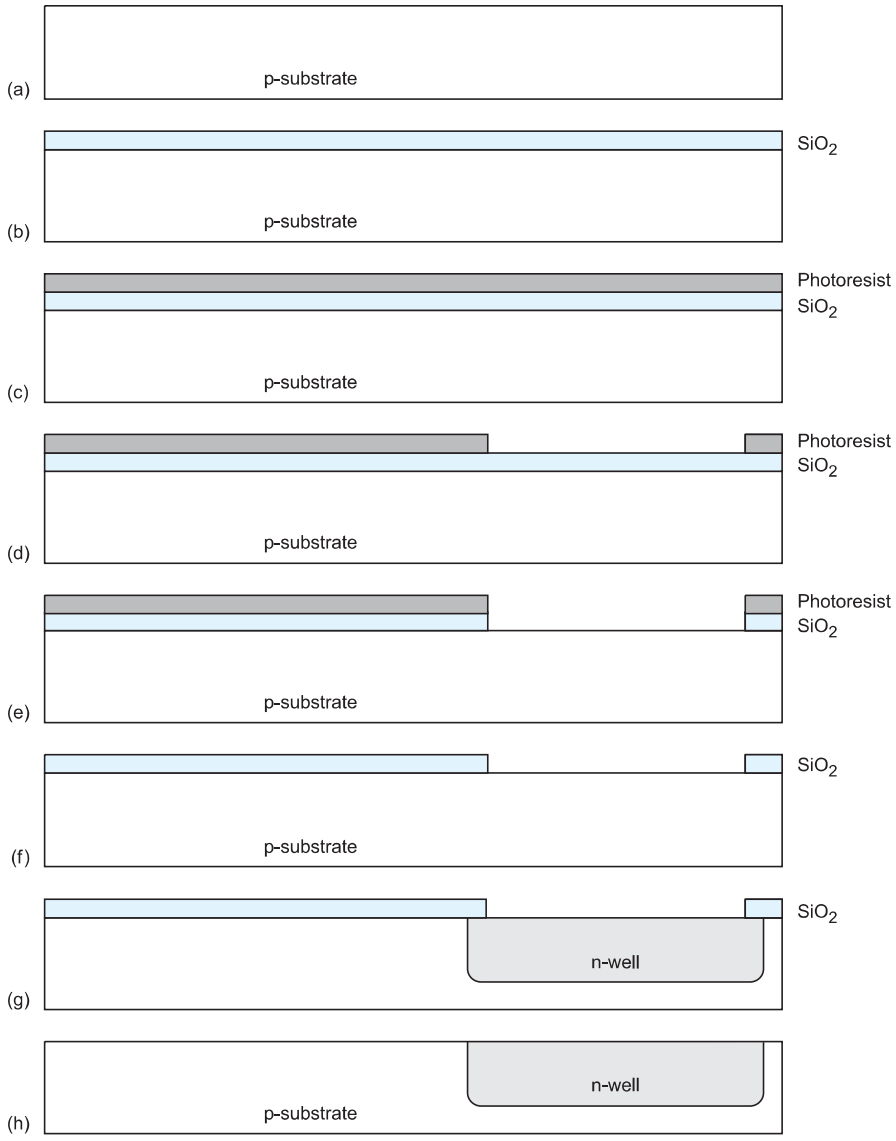


FIGURE 1.36 Cross-sections while manufacturing the n-well

[2]Engineers have experimented with many organic polymers for photoresists. In 1958, Brumford and Walker reported that Jello™ could be used for masking. They did extensive testing, observing that "various Jellos™ were evaluated with lemon giving the best result."

exposed to light is spun onto the wafer (Figure 1.36(c)). The photoresist is exposed through the n-well mask (Figure 1.35(b)) that allows light to pass through only where the well should be. The softened photoresist is removed to expose the oxide (Figure 1.36(d)). The oxide is etched with hydrofluoric acid (HF) where it is not protected by the photoresist (Figure 1.36(e)), then the remaining photoresist is stripped away using a mixture of acids called *piranha etch* (Figure 1.36(f)). The well is formed where the substrate is not covered with oxide. Two ways to add dopants are diffusion and ion implantation. In the *diffusion* process, the wafer is placed in a furnace with a gas containing the dopants. When heated, dopant atoms diffuse into the substrate. Notice how the well is wider than the hole in the oxide on account of *lateral* diffusion (Figure 1.36(g)). With *ion implantation*, dopant ions are accelerated through an electric field and blasted into the substrate. In either method, the oxide layer prevents dopant atoms from entering the substrate where no well is intended. Finally, the remaining oxide is stripped with HF to leave the bare wafer with wells in the appropriate places.

The transistor gates are formed next. These consist of polycrystalline silicon, generally called *polysilicon*, over a thin layer of oxide. The thin oxide is grown in a furnace. Then the wafer is placed in a reactor with silane gas (SiH_4) and heated again to grow the polysilicon layer through a process called *chemical vapor deposition*. The polysilicon is heavily doped to form a reasonably good conductor. The resulting cross-section is shown in Figure 1.37(a). As before, the wafer is patterned with photoresist and the polysilicon mask (Figure 1.35(c)), leaving the polysilicon gates atop the thin gate oxide (Figure 1.37(b)).

The n+ regions are introduced for the transistor active area and the well contact. As with the well, a protective layer of oxide is formed (Figure 1.37(c)) and patterned with the n-diffusion mask (Figure 1.35(d)) to expose the areas where the dopants are needed (Figure 1.37(d)). Although the n+ regions in Figure 1.37(e) are typically formed with ion implantation, they were historically diffused and thus still are often called *n-diffusion*. Notice that the polysilicon gate over the nMOS transistor blocks the diffusion so the source and drain are separated by a channel under the gate. This is called a *self-aligned* process because the source and drain of the transistor are automatically formed adjacent to the gate without the need to precisely align the masks. Finally, the protective oxide is stripped (Figure 1.37(f)).

The process is repeated for the p-diffusion mask (Figure 1.35(e)) to give the structure of Figure 1.38(a). Oxide is used for masking in the same way, and thus is not shown. The field oxide is grown to insulate the wafer from metal and patterned with the contact mask (Figure 1.35(f)) to leave contact cuts where metal should attach to diffusion or polysilicon (Figure 1.38(b)). Finally, aluminum is sputtered over the entire wafer, filling the contact cuts as well. Sputtering involves blasting aluminum into a vapor that evenly coats the wafer. The metal is patterned with the metal mask (Figure 1.35(g)) and plasma etched to remove metal everywhere except where wires should remain (Figure 1.38(c)). This completes the simple fabrication process.

Modern fabrication sequences are more elaborate because they must create complex doping profiles around the channel of the transistor and print features that are smaller than the wavelength of the light being used in lithography. However, masks for these elaborations can be automatically generated from the simple set of masks we have just examined. Modern processes also have 5–10+ layers of metal, so the metal and contact steps must be repeated for each layer. Chip manufacturing has become a commodity, and many different foundries will build designs from a basic set of masks.

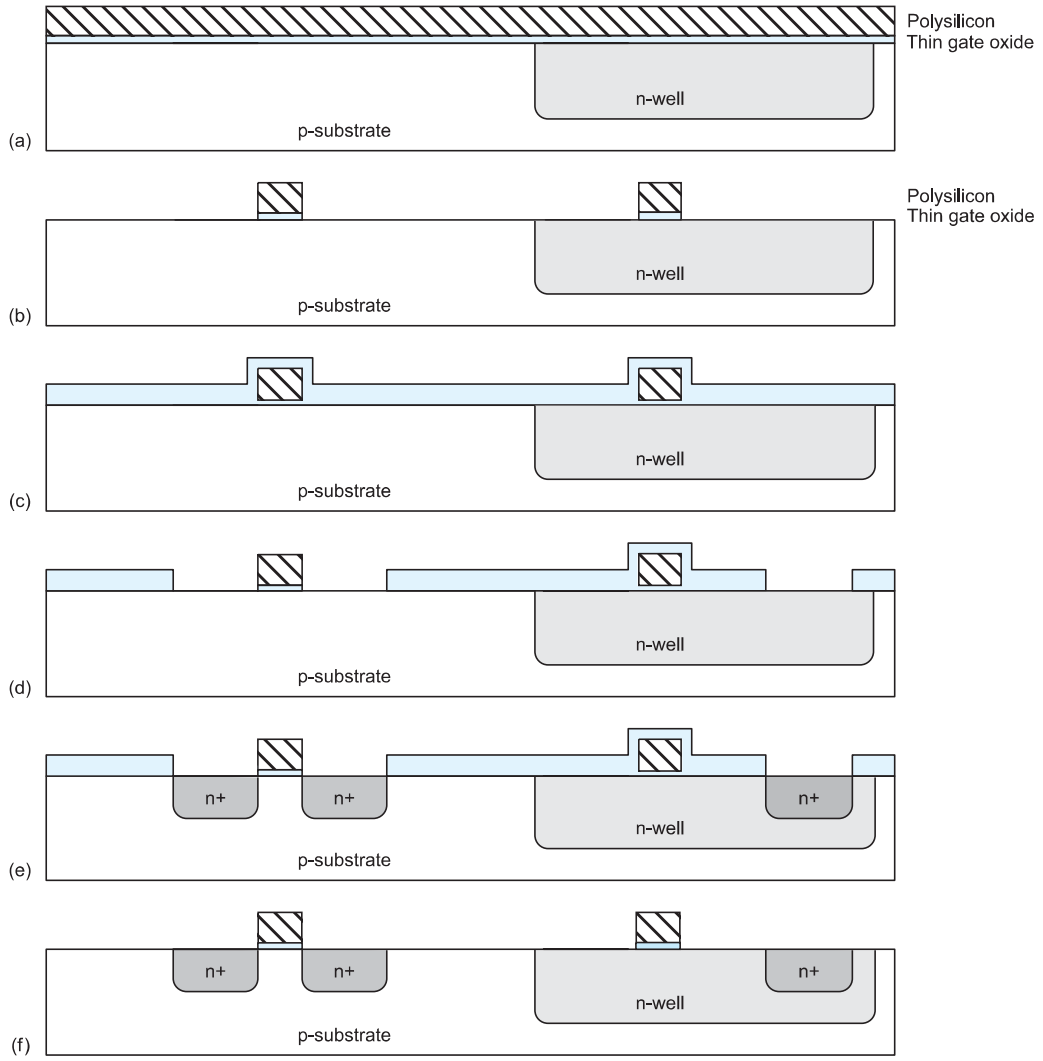


FIGURE 1.37 Cross-sections while manufacturing polysilicon and n-diffusion

1.5.3 Layout Design Rules

Layout design rules describe how small features can be and how closely they can be reliably packed in a particular manufacturing process. Industrial design rules are usually specified in microns. This makes migrating from one process to a more advanced process or a different foundry's process difficult because not all rules scale in the same way.

Universities sometimes simplify design by using scalable design rules that are conservative enough to apply to many manufacturing processes. Mead and Conway [Mead80] popularized scalable design rules based on a single parameter, λ, that characterizes the resolution of the process. λ is generally half of the minimum drawn transistor channel length. This length is the distance between the source and drain of a transistor and is set by the minimum width of a polysilicon wire. For example, a 180 nm process has a minimum polysilicon width (and hence transistor length) of 0.18 μm and uses design rules with

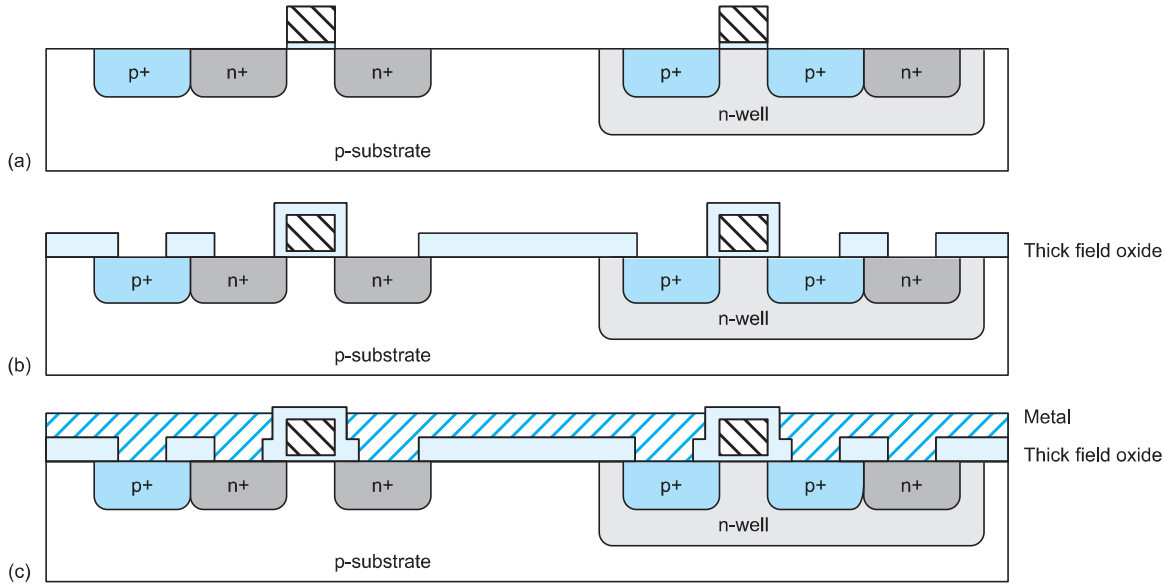


FIGURE 1.38 Cross-sections while manufacturing p-diffusion, contacts, and metal

$\lambda = 0.09\ \mu m$.[3] Lambda-based rules are necessarily conservative because they round up dimensions to an integer multiple of λ. However, they make scaling layout trivial; the same layout can be moved to a new process simply by specifying a new value of λ. This chapter will present design rules in terms of λ. The potential density advantage of micron rules is sacrificed for simplicity and easy scalability of lambda rules. Designers often describe a process by its *feature size*. Feature size refers to minimum transistor length, so λ is half the feature size.

Unfortunately, below 180 nm, design rules have become so complex and process-specific that scalable design rules are difficult to apply. However, the intuition gained from a simple set of scalable rules is still a valuable foundation for understanding the more complex rules. Chapter 3 will examine some of these process-specific rules in more detail.

The MOSIS service [Piña02] is a low-cost prototyping service that collects designs from academic, commercial, and government customers and aggregates them onto one mask set to share overhead costs and generate production volumes sufficient to interest fabrication companies. MOSIS has developed a set of scalable lambda-based design rules that covers a wide range of manufacturing processes. The rules describe the minimum width to avoid breaks in a line, minimum spacing to avoid shorts between lines, and minimum overlap to ensure that two layers completely overlap.

A conservative but easy-to-use set of design rules for layouts with two metal layers in an n-well process is as follows:

- Metal and diffusion have minimum width and spacing of 4 λ.
- Contacts are 2 $\lambda \times 2\ \lambda$ and must be surrounded by 1 λ on the layers above and below.
- Polysilicon uses a width of 2 λ.

[3]Some 180 nm lambda-based rules actually set $\lambda = 0.10\ \mu m$, then shrink the gate by 20 nm while generating masks. This keeps 180 nm gate lengths but makes all other features slightly larger.

- Polysilicon overlaps diffusion by 2 λ where a transistor is desired and has a spacing of 1 λ away where no transistor is desired.
- Polysilicon and contacts have a spacing of 3 λ from other polysilicon or contacts.
- N-well surrounds pMOS transistors by 6 λ and avoids nMOS transistors by 6 λ.

Figure 1.39 shows the basic MOSIS design rules for a process with two metal layers. Section 3.3 elaborates on these rules and compares them with industrial design rules.

In a three-level metal process, the width of the third layer is typically 6 λ and the spacing 4 λ. In general, processes with more layers often provide thicker and wider top-level metal that has a lower resistance.

Transistor dimensions are often specified by their Width/Length (W/L) ratio. For example, the nMOS transistor in Figure 1.39 formed where polysilicon crosses n-diffusion has a W/L of 4/2. In a 0.6 μm process, this corresponds to an actual width of 1.2 μm and a length of 0.6 μm. Such a minimum-width contacted transistor is often called a unit transistor.[4] pMOS transistors are often wider than nMOS transistors because holes move more slowly than electrons so the transistor has to be wider to deliver the same current. Figure 1.40(a) shows a unit inverter layout with a unit nMOS transistor and a double-sized pMOS transistor. Figure 1.40(b) shows a schematic for the inverter annotated with Width/Length for each transistor. In digital systems, transistors are typically chosen to have the minimum possible length because short-channel transistors are faster, smaller, and consume less power. Figure 1.40(c) shows a shorthand we will often use, specifying multiples of unit width and assuming minimum length.

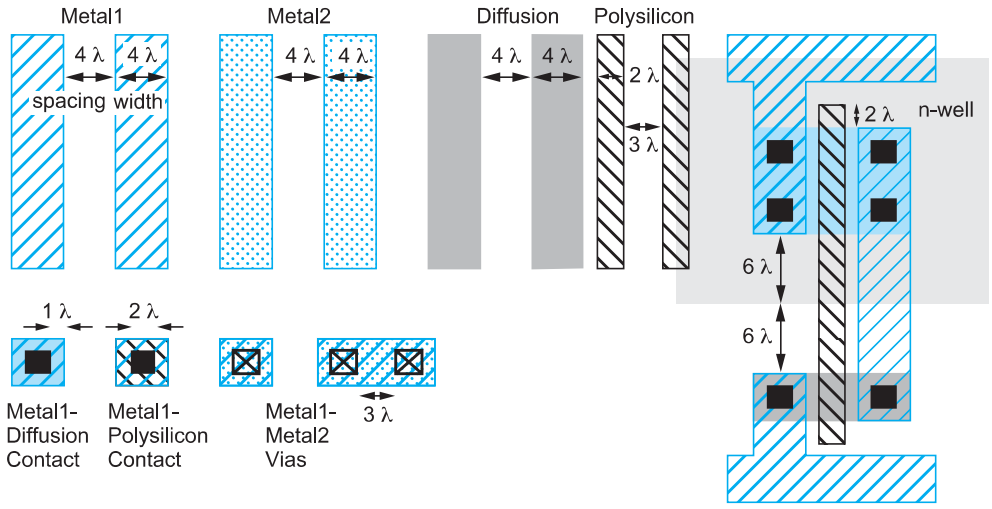


FIGURE 1.39 Simplified λ-based design rules

[4]Such small transistors in modern processes often behave slightly differently than their wider counterparts. Moreover, the transistor will not operate if either contact is damaged. Industrial designers often use a transistor wide enough for two contacts (9 λ) as the unit transistor to avoid these problems.

1.5.4 Gate Layouts

A good deal of ingenuity can be exercised and a vast amount of time wasted exploring layout topologies to minimize the size of a gate or other *cell* such as an adder or memory element. For many applications, a straightforward layout is good enough and can be automatically generated or rapidly built by hand. This section presents a simple layout style based on a "line of diffusion" rule that is commonly used for standard cells in automated layout systems. This style consists of four horizontal strips: metal ground at the bottom of the cell, n-diffusion, p-diffusion, and metal power at the top. The power and ground lines are often called *supply rails*. Polysilicon lines run vertically to form transistor gates. Metal wires within the cell connect the transistors appropriately.

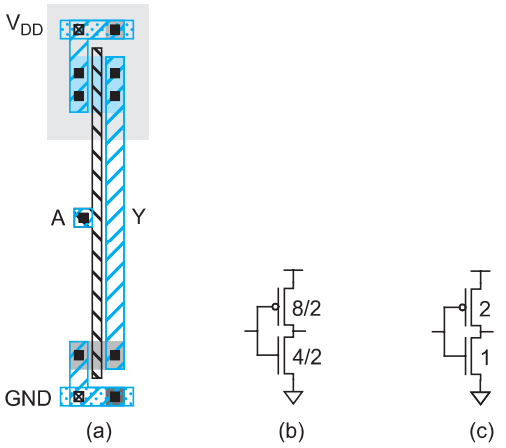


FIGURE 1.40 Inverter with dimensions labeled

Figure 1.41(a) shows such a layout for an inverter. The input *A* can be connected from the top, bottom, or left in polysilicon. The output *Y* is available at the right side of the cell in metal. Recall that the p-substrate and n-well must be tied to ground and power, respectively. Figure 1.41(b) shows the same inverter with well and substrate taps placed under the power and ground rails, respectively. Figure 1.42 shows a 3-input NAND gate. Notice how the nMOS transistors are connected in series while the pMOS transistors are connected in parallel. Power and ground extend 2 λ on each side so if two gates were abutted the contents would be separated by 4 λ, satisfying design rules. The height of the cell is 36 λ, or 40 λ if the 4 λ space between the cell and another wire above it is counted. All these examples use transistors of width 4 λ. Choice of transistor width is addressed further in Chapters 4–5 and cell layout styles are discussed in Section 14.7.

These cells were designed such that the gate connections are made from the top or bottom in polysilicon. In contemporary standard cells, polysilicon is generally not used as a routing layer so the cell must allow metal2 to metal1 and metal1 to polysilicon contacts

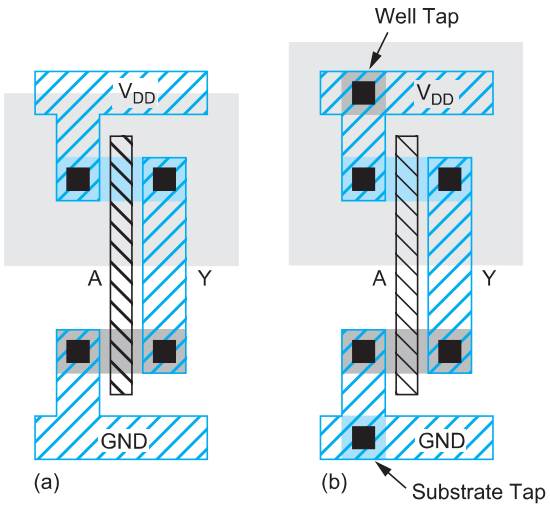


FIGURE 1.41 Inverter cell layout

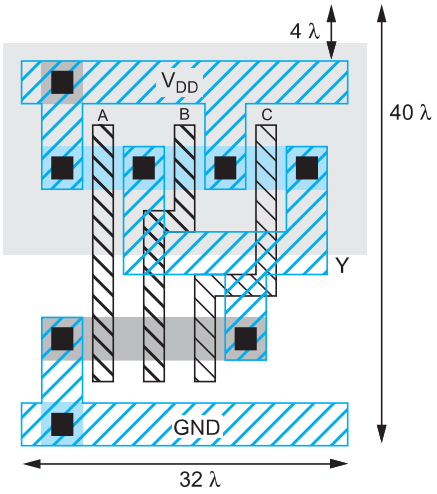


FIGURE 1.42 3-input NAND standard cell gate layouts

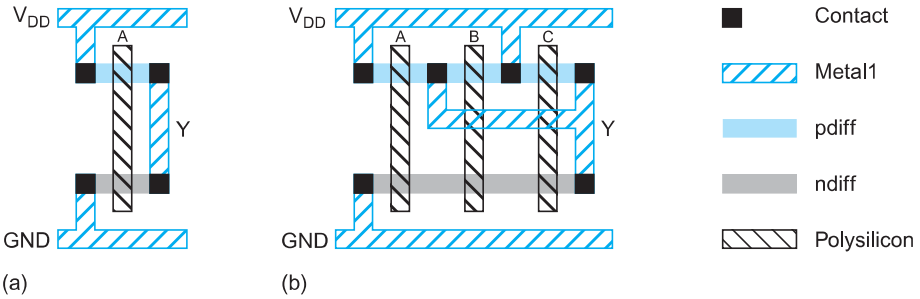


FIGURE 1.43 Stick diagrams of inverter and 3-input NAND gate. Color version on inside front cover.

to each gate. While this increases the size of the cell, it allows free access to all terminals on metal routing layers.

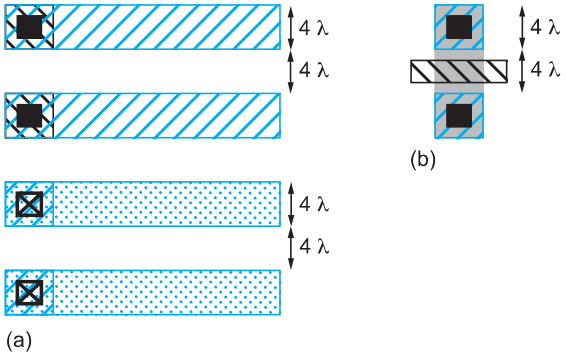


FIGURE 1.44 Pitch of routing tracks

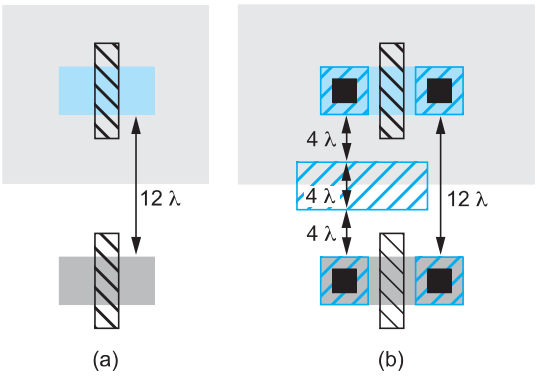


FIGURE 1.45 Spacing between nMOS and pMOS transistors

1.5.5 Stick Diagrams

Because layout is time-consuming, designers need fast ways to plan cells and estimate area before committing to a full layout. *Stick diagrams* are easy to draw because they do not need to be drawn to scale. Figure 1.43 and the inside front cover show stick diagrams for an inverter and a 3-input NAND gate. While this book uses stipple patterns, layout designers use dry-erase markers or colored pencils.

With practice, it is easy to estimate the area of a layout from the corresponding stick diagram even though the diagram is not to scale. Although schematics focus on transistors, layout area is usually determined by the metal wires. Transistors are merely widgets that fit under the wires. We define a *routing track* as enough space to place a wire and the required spacing to the next wire. If our wires have a width of 4 λ and a spacing of 4 λ to the next wire, the track *pitch* is 8 λ, as shown in Figure 1.44(a). This pitch also leaves room for a transistor to be placed between the wires (Figure 1.44(b)). Therefore, it is reasonable to estimate the height and width of a cell by counting the number of metal tracks and multiplying by 8 λ. A slight complication is the required spacing of 12 λ between nMOS and pMOS transistors set by the well, as shown in Figure 1.45(a). This space can be occupied by an additional track of wire, shown in Figure 1.45(b). Therefore, an extra track must be allocated between nMOS and pMOS transistors regardless of whether wire is actually used in that track. Figure 1.46 shows how to count tracks to estimate the size of a 3-input NAND. There are four vertical wire tracks, multiplied by 8 λ per track to give a cell width of 32 λ. There are five horizontal tracks, giving a cell height of 40 λ. Even though the horizontal tracks are not drawn to scale, they are still easy to count. Figure 1.42

shows that the actual NAND gate layout matches the dimensions predicted by the stick diagram. If transistors are wider than 4 λ, the extra width must be factored into the area estimate. Of course, these estimates are oversimplifications of the complete design rules and a trial layout should be performed for truly critical cells.

Example 1.3

Sketch a stick diagram for a CMOS gate computing $Y = \overline{(A + B + C) \cdot D}$ (see Figure 1.18) and estimate the cell width and height.

SOLUTION: Figure 1.47 shows a stick diagram. Counting horizontal and vertical pitches gives an estimated cell size of 40 by 48 λ.

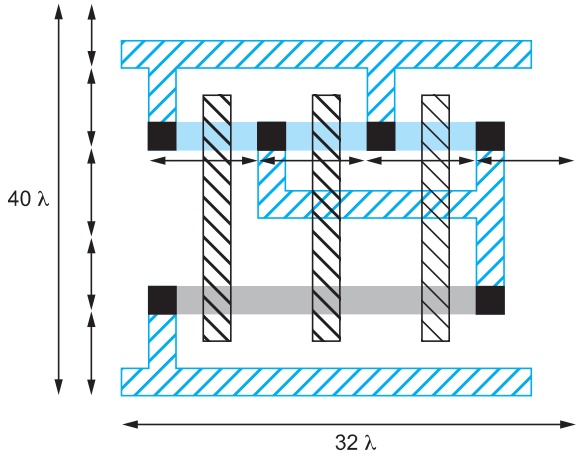


FIGURE 1.46 3-input NAND gate area estimation

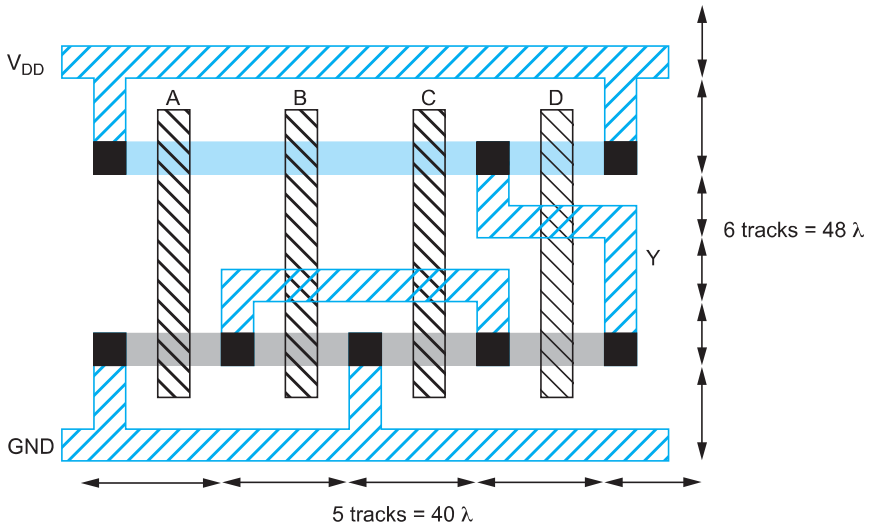


FIGURE 1.47 CMOS compound gate for function $Y = \overline{(A + B + C) \cdot D}$

1.6 Design Partitioning

By this point, you know that MOS transistors behave as voltage-controlled switches. You know how to build logic gates out of transistors. And you know how transistors are fabricated and how to draw a layout that specifies how transistors should be placed and connected together. You know enough to start building your own simple chips.

The greatest challenge in modern VLSI design is not in designing the individual transistors but rather in managing system complexity. Modern *System-On-Chip* (SOC) designs combine memories, processors, high-speed I/O interfaces, and dedicated application-specific logic on a single chip. They use hundreds of millions or billions of transistors and cost tens of millions of dollars (or more) to design. The implementation

must be divided among large teams of engineers and each engineer must be highly productive. If the implementation is too rigidly partitioned, each block can be optimized without regard to its neighbors, leading to poor system results. Conversely, if every task is interdependent with every other task, design will progress too slowly. Design managers face the challenge of choosing a suitable trade-off between these extremes. There is no substitute for practical experience in making these choices, and talented engineers who have experience with multiple designs are very important to the success of a large project. Design proceeds through multiple levels of abstraction, hiding details until they become necessary. The practice of *structured design*, which is also used in large software projects, uses the principles of hierarchy, regularity, modularity, and locality to manage the complexity.

1.6.1 Design Abstractions

Digital VLSI design is often partitioned into five levels of abstractions: *architecture* design, *microarchitecture* design, *logic* design, *circuit* design, and *physical* design. Architecture describes the functions of the system. For example, the x86 microprocessor architecture specifies the instruction set, register set, and memory model. Microarchitecture describes how the architecture is partitioned into registers and functional units. The 80386, 80486, Pentium, Pentium II, Pentium III, Pentium 4, Core, Core 2, Atom, Cyrix MII, AMD Athlon, and Phenom are all microarchitectures offering different performance / transistor count / power trade-offs for the x86 architecture. Logic describes how functional units are constructed. For example, various logic designs for a 32-bit adder in the x86 integer unit include ripple carry, carry lookahead, and carry select. Circuit design describes how transistors are used to implement the logic. For example, a carry lookahead adder can use static CMOS circuits, domino circuits, or pass transistors. The circuits can be tailored to emphasize high performance or low power. Physical design describes the layout of the chip. Analog and RF VLSI design involves the same steps but with different layers of abstraction.

These elements are inherently interdependent and all influence each of the design objectives. For example, choices of microarchitecture and logic are strongly dependent on the number of transistors that can be placed on the chip, which depends on the physical design and process technology. Similarly, innovative circuit design that reduces a cache access from two cycles to one can influence which microarchitecture is most desirable. The choice of clock frequency depends on a complex interplay of microarchitecture and logic, circuit design, and physical design. Deeper pipelines allow higher frequencies but consume more power and lead to greater performance penalties when operations early in the pipeline are dependent on those late in the pipeline. Many functions have various logic and circuit designs trading speed for area, power, and design effort. Custom physical design allows more compact, faster circuits and lower manufacturing costs, but involves an enormous labor cost. Automatic layout with CAD systems reduces the labor and achieves faster times to market.

To deal with these interdependencies, microarchitecture, logic, circuit, and physical design must occur, at least in part, in parallel. Microarchitects depend on circuit and physical design studies to understand the cost of proposed microarchitectural features. Engineers are sometimes categorized as “short and fat” or “tall and skinny” (nothing personal, we assure you!). Tall, skinny engineers understand something about a broad range of topics. Short, fat engineers understand a large amount about a narrow field. Digital VLSI design favors the tall, skinny engineer who can evaluate how choices in one part of the system impact other parts of the system.

1.6.2 Structured Design

Hierarchy is a critical tool for managing complex designs. A large system can be partitioned hierarchically into multiple *cores*. Each core is built from various *units*. Each unit in turn is composed of multiple *functional blocks*.[5] These blocks in turn are built from *cells*, which ultimately are constructed from transistors. The system can be more easily understood at the top level by viewing components as black boxes with well-defined interfaces and functions rather than looking at each individual transistor. Logic, circuit, and physical views of the design should share the same hierarchy for ease of verification. A design hierarchy can be viewed as a tree structure with the overall chip as the *root* and the primitive cells as *leafs*.

Regularity aids the management of design complexity by designing the minimum number of different blocks. Once a block is designed and verified, it can be reused in many places. *Modularity* requires that the blocks have well-defined interfaces to avoid unanticipated interactions. *Locality* involves keeping information where it is used, physically and temporally. Structured design is discussed further in Section 14.2.

1.6.3 Behavioral, Structural, and Physical Domains

An alternative way of viewing design partitioning is shown with the Y-chart shown in Figure 1.48 [Gajski83, Kang03]. The radial lines on the Y-chart represent three distinct design domains: behavioral, structural, and physical. These domains can be used to describe the design of almost any artifact and thus form a general taxonomy for describing

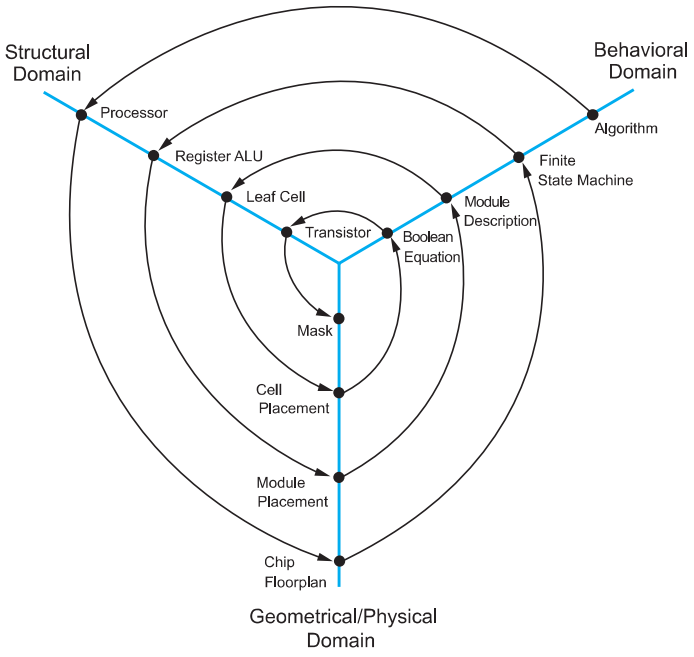


FIGURE 1.48 Y Diagram (Reproduced from [Kang03] with permission of The McGraw-Hill Companies.)

[5]Some designers refer to both units and functional blocks as *modules*.

the design process. Within each domain there are a number of levels of design abstraction that start at a very high level and descend eventually to the individual elements that need to be aggregated to yield the top level function (i.e., transistors in the case of chip design).

The behavioral domain describes what a particular system does. For instance, at the highest level we might specify a telephone touch-tone generator. This behavior can be successively refined to more precisely describe what needs to be done in order to build the tone generator (i.e., the frequencies desired, output levels, distortion allowed, etc.).

At each abstraction level, a corresponding structural description can be developed. The structural domain describes the interconnection of modules necessary to achieve a particular behavior. For instance, at the highest level, the touch-tone generator might consist of a keypad, a tone generator chip, an audio amplifier, a battery, and a speaker. Eventually at lower levels of abstraction, the individual gate and then transistor connections required to build the tone generator are described.

For each level of abstraction, the physical domain description explains how to physically construct that level of abstraction. At high levels, this might consist of an engineering drawing showing how to put together the keypad, tone generator chip, battery, and speaker in the associated housing. At the top chip level, this might consist of a floorplan, and at lower levels, the actual geometry of individual transistors.

The design process can be viewed as making transformations from one domain to another while maintaining the equivalency of the domains. Behavioral descriptions are transformed to structural descriptions, which in turn are transformed to physical descriptions. These transformations can be manual or automatic. In either case, it is normal design practice to verify the transformation of one domain to the other. This ensures that the design intent is carried across the domain boundaries. Hierarchically specifying each domain at successively detailed levels of abstraction allows us to design very large systems.

The reason for strictly describing the domains and levels of abstraction is to define a precise design process in which the final function of the system can be traced all the way back to the initial behavioral description. In an ideal flow, there should be no opportunity to produce an incorrect design. If anomalies arise, the design process is corrected so that those anomalies will not reoccur in the future. A designer should acquire a rigid discipline with respect to the design process, and be aware of each transformation and how and why it is failproof. Normally, these steps are fully automated in a modern design process, but it is important to be aware of the basis for these steps in order to debug them if they go astray.

The Y diagram can be used to illustrate each domain and the transformations between domains at varying levels of design abstraction. As the design process winds its way from the outer to inner rings, it proceeds from higher to lower levels of abstraction and hierarchy.

Most of the remainder of this chapter is a case study in the design of a simple microprocessor to illustrate the various aspects of VLSI design applied to a nontrivial system. We begin by describing the architecture and microarchitecture of the processor. We then consider logic design and discuss hardware description languages. The processor is built with static CMOS circuits, which we examined in Section 1.4; transistor-level design and netlist formats are discussed. We continue exploring the physical design of the processor including floorplanning and area estimation. Design verification is critically important and happens at each level of the hierarchy for each element of the design. Finally, the layout is converted into masks so the chip can be manufactured, packaged, and tested.

1.7 Example: A Simple MIPS Microprocessor

We consider an 8-bit subset of the MIPS microprocessor architecture [Patterson04, Harris07] because it is widely studied and is relatively simple, yet still large enough to illustrate hierarchical design. This section describes the architecture and the multicycle microarchitecture we will be implementing. If you are not familiar with computer architecture, you can regard the MIPS processor as a black box and skip to Section 1.8.

A set of laboratory exercises is available at `www.cmosvlsi.com` in which you can learn VLSI design by building the microprocessor yourself using a free open-source CAD tool called *Electric* or with commercial design tools from Cadence and Synopsys.

1.7.1 MIPS Architecture

The MIPS32 architecture is a simple 32-bit RISC architecture with relatively few idiosyncrasies. Our subset of the architecture uses 32-bit instruction encodings but only eight 8-bit general-purpose registers named `$0`–`$7`. We also use an 8-bit program counter (PC). Register `$0` is hardwired to contain the number 0. The instructions are `ADD`, `SUB`, `AND`, `OR`, `SLT`, `ADDI`, `BEQ`, `J`, `LB`, and `SB`.

The function and encoding of each instruction is given in Table 1.7. Each instruction is encoded using one of three templates: R, I, and J. R-type instructions (*register*-based) are used for arithmetic and specify two source registers and a destination register. I-type instructions are used when a 16-bit constant (also known as an *immediate*) and two registers must be specified. J-type instructions (*jumps*) dedicate most of the instruction word to a 26-bit jump destination. The format of each encoding is defined in Figure 1.49. The six most significant bits of all formats are the operation code (`op`). R-type instructions all share `op` = 000000 and use six more `funct` bits to differentiate the functions.

TABLE 1.7 MIPS instruction set (subset supported)

Instruction	Function		Encoding	op	funct
`add $1, $2, $3`	addition:	`$1 <- $2 + $3`	R	000000	100000
`sub $1, $2, $3`	subtraction:	`$1 <- $2 – $3`	R	000000	100010
`and $1, $2, $3`	bitwise and:	`$1 <- $2 and $3`	R	000000	100100
`or $1, $2, $3`	bitwise or:	`$1 <- $2 or $3`	R	000000	100101
`slt $1, $2, $3`	set less than:	`$1 <- 1 if $2 < $3` `$1 <- 0 otherwise`	R	000000	101010
`addi $1, $2, imm`	add immediate:	`$1 <- $2 + imm`	I	001000	n/a
`beq $1, $2, imm`	branch if equal:	`PC <- PC + imm × 4`[a]	I	000100	n/a
`j destination`	jump:	`PC <- destination`[a]	J	000010	n/a
`lb $1, imm($2)`	load byte:	`$1 <- mem[$2 + imm]`	I	100000	n/a
`sb $1, imm($2)`	store byte:	`mem[$2 + imm] <- $1`	I	101000	n/a

a. Technically, MIPS addresses specify bytes. Instructions require a 4-byte word and must begin at addresses that are a multiple of four. To most effectively use instruction bits in the full 32-bit MIPS architecture, branch and jump constants are specified in words and must be multiplied by four (shifted left 2 bits) to be converted to byte addresses.

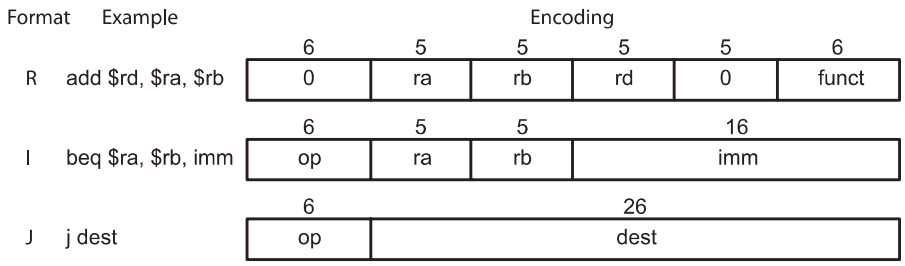


FIGURE 1.49 Instruction encoding formats

We can write programs for the MIPS processor in *assembly language*, where each line of the program contains one instruction such as `ADD` or `BEQ`. However, the MIPS hardware ultimately must read the program as a series of 32-bit numbers called *machine language*. An *assembler* automates the tedious process of translating from assembly language to machine language using the encodings defined in Table 1.7 and Figure 1.49. Writing nontrivial programs in assembly language is also tedious, so programmers usually work in a *high-level language* such as C or Java. A *compiler* translates a program from high-level language *source code* into the appropriate machine language *object code*.

Example 1.4

Figure 1.50 shows a simple C program that computes the nth Fibonacci number f_n defined recursively for $n > 0$ as $f_n = f_{n-1} + f_{n-2}, f_{-1} = -1, f_0 = 1$. Translate the program into MIPS assembly language and machine language.

SOLUTION: Figure 1.51 gives a commented assembly language program. Figure 1.52 translates the assembly language to machine language.

```
int fib(void)
{
   int n = 8;             /* compute nth Fibonacci number */
   int f1 = 1, f2 = -1;   /* last two Fibonacci numbers */

   while (n != 0) {       /* count down to n = 0 */
     f1 = f1 + f2;
     f2 = f1 - f2;
     n = n - 1;
   }
   return f1;
```

FIGURE 1.50 C Code for Fibonacci program

1.7.2 Multicycle MIPS Microarchitecture

We will implement the multicycle MIPS microarchitecture given in Chapter 5 of [Patterson04] and Chapter 7 of [Harris07] modified to process 8-bit data. The microarchitecture is illustrated in Figure 1.53. Light lines indicate individual signals while heavy

```
# fib.asm
# Register usage: $3: n $4: f1 $5: f2
# return value written to address 255
fib:  addi $3, $0, 8      # initialize n=8
      addi $4, $0, 1      # initialize f1 = 1
      addi $5, $0, -1     # initialize f2 = -1
loop: beq $3, $0, end     # Done with loop if n = 0
      add $4, $4, $5      # f1 = f1 + f2
      sub $5, $4, $5      # f2 = f1 - f2
      addi $3, $3, -1     # n = n - 1
      j loop              # repeat until done
end:  sb $4, 255($0)      # store result in address 255
```

FIGURE 1.51 Assembly language code for Fibonacci program

Instruction	Binary Encoding	Hexadecimal Encoding
addi $3, $0, 8	001000 00000 00011 0000000000001000	20030008
addi $4, $0, 1	001000 00000 00100 0000000000000001	20040001
addi $5, $0, -1	001000 00000 00101 1111111111111111	2005ffff
beq $3, $0, end	000100 00011 00000 0000000000000100	10600004
add $4, $4, $5	000000 00100 00101 00100 00000 100000	00852020
sub $5, $4, $5	000000 00100 00101 00101 00000 100010	00852822
addi $3, $3, -1	001000 00011 00011 1111111111111111	2063ffff
j loop	000010 00000000000000000000000011	08000003
sb $4, 255($0)	101000 00000 00100 0000000011111111	a00400ff

FIGURE 1.52 Machine language code for Fibonacci program

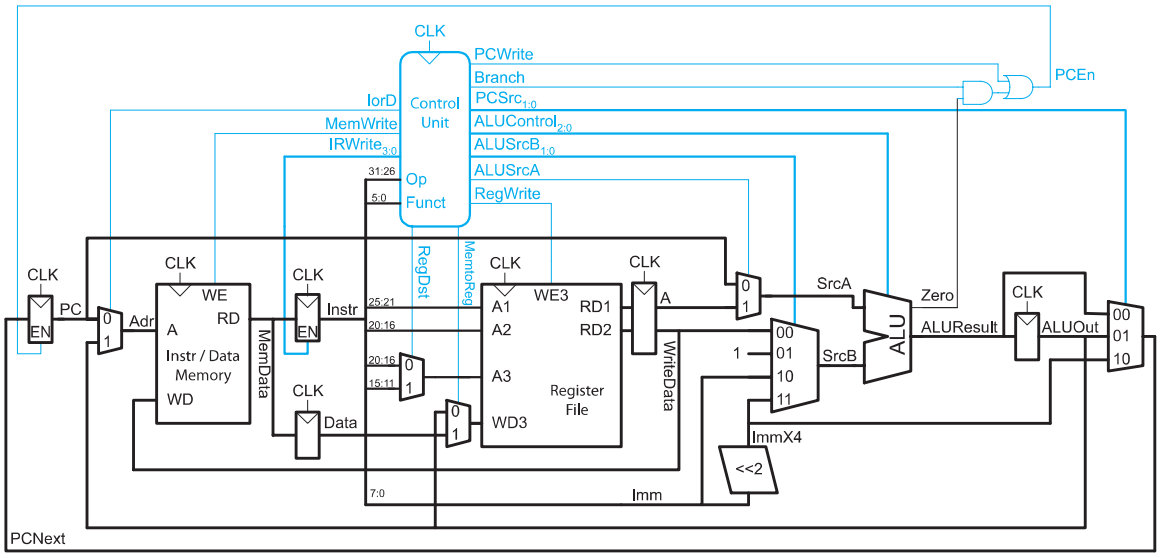


FIGURE 1.53 Multicycle MIPS microarchitecture. Adapted from [Patterson04] and [Harris07] with permission from Elsevier.

lines indicate busses. The control logic and signals are highlighted in blue while the datapath is shown in black. Control signals generally drive multiplexer select signals and register enables to tell the datapath how to execute an instruction.

Instruction execution generally flows from left to right. The program counter (PC) specifies the address of the instruction. The instruction is loaded 1 byte at a time over four cycles from an off-chip memory into the 32-bit instruction register (IR). The `Op` field (bits 31:26 of the instruction) is sent to the controller, which sequences the datapath through the correct operations to execute the instruction. For example, in an `ADD` instruction, the two source registers are read from the register file into temporary registers A and B. On the next cycle, the aludec unit commands the Arithmetic/Logic Unit (ALU) to add the inputs. The result is captured in the `ALUOut` register. On the third cycle, the result is written back to the appropriate destination register in the register file.

The controller contains a finite state machine (FSM) that generates multiplexer select signals and register enables to sequence the datapath. A state transition diagram for the FSM is shown in Figure 1.54. As discussed, the first four states fetch the instruction from

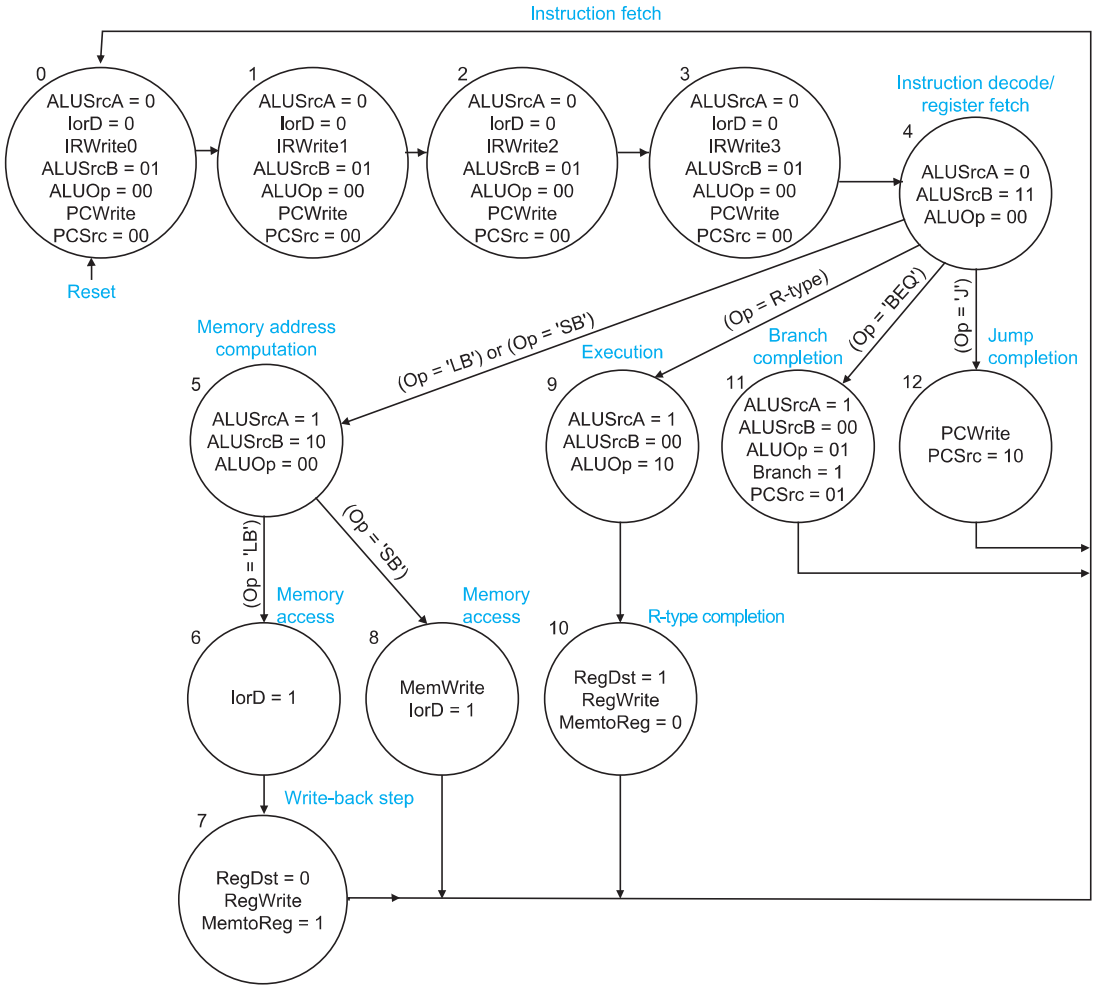


FIGURE 1.54 Multicycle MIPS control FSM (Adapted from [Patterson04] and [Harris07] with permission from Elsevier.)

memory. The FSM then is dispatched based on `Op` to execute the particular instruction. The FSM states for `ADDI` are missing and left as an exercise for the reader.

Observe that the FSM produces a 2-bit `ALUOp` output. The ALU decoder unit in the controller uses combinational logic to compute a 3-bit `ALUControl` signal from the `ALUOp` and `Funct` fields, as specified in Table 1.8. `ALUControl` drives multiplexers in the ALU to select the appropriate computation.

TABLE 1.8 ALUControl determination

ALUOp	Funct	ALUControl	Meaning
00	x	010	ADD
01	x	110	SUB
10	100000	010	ADD
10	100010	110	SUB
10	100100	000	AND
10	100101	001	OR
10	101010	111	SLT
11	x	x	undefined

Example 1.5

Referring to Figures 1.53 and 1.54, explain how the MIPS processor fetches and executes the `SUB` instruction.

SOLUTION: The first step is to fetch the 32-bit instruction. This takes four cycles because the instruction must come over an 8-bit memory interface. On each cycle, we want to fetch a byte from the address in memory specified by the program counter, then increment the program counter by one to point to the next byte.

The fetch is performed by states 0–3 of the FSM in Figure 1.54. Let us start with state 0. The program counter (PC) contains the address of the first byte of the instruction. The controller must select `IorD` = 0 so that the multiplexer sends this address to the memory. `MemRead` must also be asserted so the memory reads the byte onto the `MemData` bus. Finally, `IRWrite0` should be asserted to enable writing `memdata` into the least significant byte of the instruction register (IR).

Meanwhile, we need to increment the program counter. We can do this with the ALU by specifying PC as one input, 1 as the other input, and `ADD` as the operation. To select PC as the first input, `ALUSrcA` = 0. To select 1 as the other input, `ALUSrcB` = 01. To perform an addition, `ALUOp` = 00, according to Table 1.8. To write this result back into the program counter at the end of the cycle, `PCSrc` = 00 and `PCEn` = 1 (done by setting `PCWrite` = 1).

All of these control signals are indicated in state 0 of Figure 1.54. The other register enables are assumed to be 0 if not explicitly asserted and the other multiplexer selects are don't cares. The next three states are identical except that they write bytes 1, 2, and 3 of the IR, respectively.

The next step is to read the source registers, done in state 4. The two source registers are specified in bits 25:21 and 20:16 of the IR. The register file reads these registers and puts the values into the A and B registers. No control signals are necessary for `SUB` (although state 4 performs a branch address computation in case the instruction is `BEQ`).

The next step is to perform the subtraction. Based on the `Op` field (IR bits 31:26), the FSM jumps to state 9 because `SUB` is an R-type instruction. The two source registers are selected as input to the ALU by setting `ALUSrcA` = 1 and `ALUSrcB` = 00. Choosing `ALUOp` = 10 directs the ALU Control decoder to select the `ALUControl` signal as 110, subtraction. Other R-type instructions are executed identically except that the decoder receives a different `Funct` code (IR bits 5:0) and thus generates a different `ALUControl` signal. The result is placed in the ALUOut register.

Finally, the result must be written back to the register file in state 10. The data comes from the ALUOut register so `MemtoReg` = 0. The destination register is specified in bits 15:11 of the instruction so `RegDst` = 1. `RegWrite` must be asserted to perform the write. Then, the control FSM returns to state 0 to fetch the next instruction.

1.8 Logic Design

We begin the logic design by defining the top-level chip interface and block diagram. We then hierarchically decompose the units until we reach leaf cells. We specify the logic with a Hardware Description Language (HDL), which provides a higher level of abstraction than schematics or layout. This code is often called the Register Transfer Level (RTL) description.

1.8.1 Top-Level Interfaces

The top-level inputs and outputs are listed in Table 1.9. This example uses a two-phase clocking system to avoid hold-time problems. `Reset` initializes the PC to 0 and the control FSM to the start state.

TABLE 1.9 Top-level inputs and outputs

Inputs	Outputs
`ph1`	`MemWrite`
`ph2`	`Adr[7:0]`
`reset`	`WriteData[7:0]`
`MemData[7:0]`	

The remainder of the signals are used for an 8-bit memory interface (assuming the memory is located off chip). The processor sends an 8-bit address `Adr` and optionally asserts `MemWrite`. On a read cycle, the memory returns a value on the `MemData` lines while on a write cycle, the memory accepts input from `WriteData`. In many systems, `MemData` and `WriteData` can be combined onto a single bidirectional bus, but for this example we preserve the interface of Figure 1.53. Figure 1.55 shows a simple computer system built from the MIPS processor, external memory, reset switch, and clock generator.

1.8.2 Block Diagrams

The chip is partitioned into two top-level units: the controller and datapath, as shown in the block diagram in Figure 1.56. The controller comprises the control FSM, the ALU decoder, and the two gates used to compute `PCEn`. The ALU decoder consists of combina-

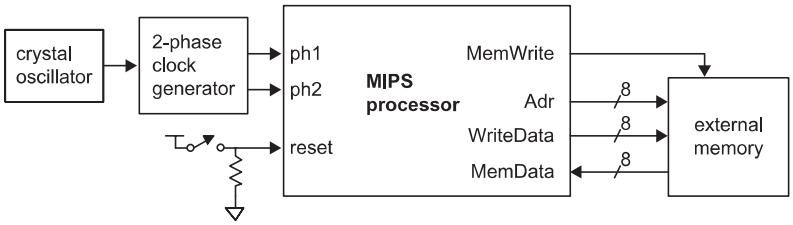


FIGURE 1.55 MIPS computer system

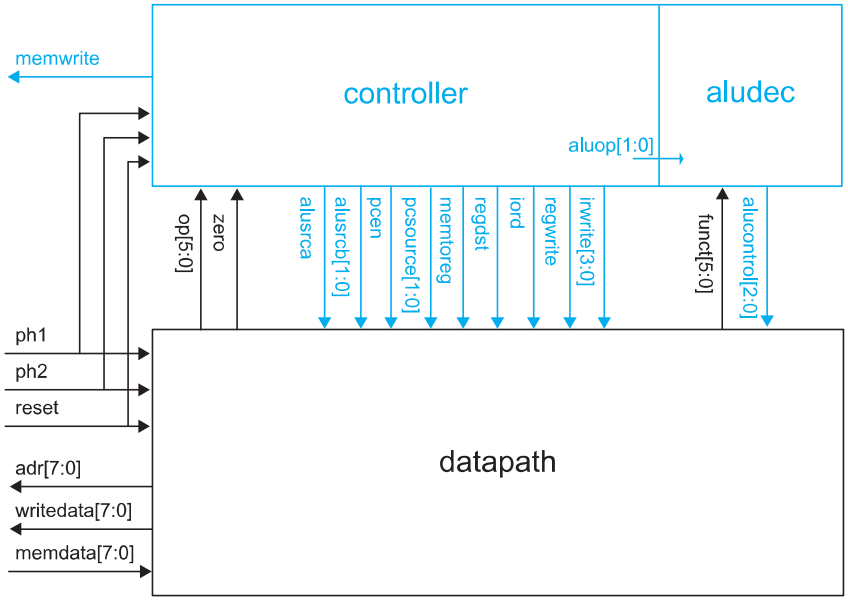


FIGURE 1.56 Top-level MIPS block diagram

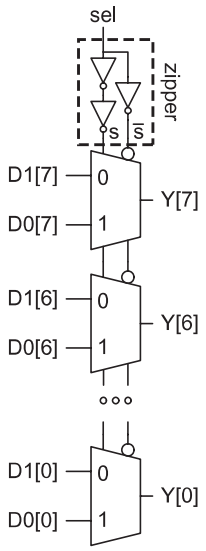


FIGURE 1.57 8-bit 2:1 multiplexer wordslice

tional logic to determine `ALUControl`. The 8-bit datapath contains the remainder of the chip. It can be viewed as a collection of wordslices or bitslices. A *wordslice* is a column containing an 8-bit flip-flop, adder, multiplexer, or other element. For example, Figure 1.57 shows a wordslice for an 8-bit 2:1 multiplexer. It contains eight individual 2:1 multiplexers, along with a *zipper* containing a buffer and inverter to drive the true and complementary select signals to all eight multiplexers.[6] Factoring these drivers out into the zipper saves space as compared to putting inverters in each multiplexer. Alternatively, the datapath can be viewed as eight rows of *bitslices*. Each bitslice has one bit of each component, along with the horizontal wires connecting the bits together.

The chip partitioning is influenced by the intended physical design. The datapath contains most of the transistors and is very regular in structure. We can achieve high density with moderate design effort by handcrafting each wordslice or bitslice and tiling the

[6] In this example, the zipper is shown at the top of the wordslice. In wider datapaths, the zipper is sometimes placed in the middle of the wordslice so that it drives shorter wires. The name comes from the way the layout resembles a plaid sweatshirt with a zipper down the middle.

circuits together. Building datapaths using wordslices is usually easier because certain structures, such as the zero detection circuit in the ALU, are not identical in each bitslice. However, thinking about bitslices is a valuable way to plan the wiring across the datapath. The controller has much less structure. It is tedious to translate an FSM into gates by hand, and in a new design, the controller is the most likely portion to have bugs and last-minute changes. Therefore, we will specify the controller more abstractly with a hardware description language and automatically generate it using synthesis and place & route tools or a programmable logic array (PLA).

1.8.3 Hierarchy

The best way to design complex systems is to decompose them into simpler pieces. Figure 1.58 shows part of the design hierarchy for the MIPS processor. The controller contains the controller_pla and aludec, which in turn is built from a library of standard cells such as NANDs, NORs, and inverters. The datapath is composed of 8-bit wordslices, each of which also is typically built from standard cells such as adders, register file bits, multiplexers, and flip-flops. Some of these cells are reused in multiple places.

The design hierarchy does not necessarily have to be identical in the logic, circuit, and physical designs. For example, in the logic view, a memory may be best treated as a black box, while in the circuit implementation, it may have a decoder, cell array, column multiplexers, and so forth. Different hierarchies complicate verification, however, because they must be *flattened* until the point that they agree. As a matter of practice, it is best to make logic, circuit, and physical design hierarchies agree as far as possible.

1.8.4 Hardware Description Languages

Designers need rapid feedback on whether a logic design is reasonable. Translating block diagrams and FSM state transition diagrams into circuit schematics is time-consuming and prone to error; before going through this entire process it is wise to know if the top-level design has major bugs that will require complete redesign. HDLs provide a way to specify the design at a higher level of abstraction to raise designer productivity. They were originally intended for documentation and simulation, but are now used to synthesize gates directly from the HDL.

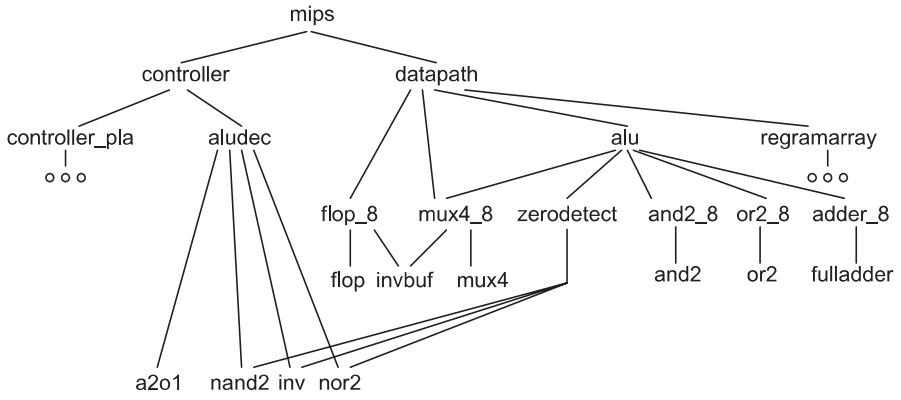


FIGURE 1.58 MIPS design hierarchy

The two most popular HDLs are *Verilog* and *VHDL*. Verilog was developed by Advanced Integrated Design Systems (later renamed Gateway Design Automation) in 1984 and became a *de facto* industry open standard by 1991. In 2005, the SystemVerilog extensions were standardized, and some of these features are used in this book. VHDL, which stands for VHSIC Hardware Description Language, where VHSIC in turn was a Department of Defense project on Very High Speed Integrated Circuits, was developed by committee under government sponsorship. As one might expect from their pedigrees, Verilog is less verbose and closer in syntax to C, while VHDL supports some abstractions useful for large team projects. Many Silicon Valley companies use Verilog while defense and telecommunications companies often use VHDL. Neither language offers a decisive advantage over the other so the industry is saddled with supporting both. Appendix A offers side-by-side tutorials on Verilog and VHDL. Examples in this book are given in Verilog for the sake of brevity.

When coding in an HDL, it is important to remember that you are specifying hardware that operates in parallel rather than software that executes in sequence. There are two general coding styles. *Structural* HDL specifies how a cell is composed of other cells or primitive gates and transistors. *Behavioral* HDL specifies what a cell does.

A *logic simulator* simulates HDL code; it can report whether results match expectations, and can display waveforms to help debug discrepancies. A *logic synthesis* tool is similar to a compiler for hardware: it maps HDL code onto a *library* of gates called *standard cells* to minimize area while meeting some timing constraints. Only a subset of HDL constructs are synthesizable; this subset is emphasized in the appendix. For example, file I/O commands used in testbenches are obviously not synthesizable. Logic synthesis generally produces circuits that are neither as dense nor as fast as those handcrafted by a skilled designer. Nevertheless, integrated circuit processes are now so advanced that synthesized circuits are good enough for the great majority of application-specific integrated circuits (ASICs) built today. Layout may be automatically generated using place & route tools.

Verilog and VHDL models for the MIPS processor are listed in Appendix A.12. In Verilog, each cell is called a *module*. The inputs and outputs are declared much as in a C program and bit widths are given for busses. Internal signals must also be declared in a way analogous to local variables. The processor is described hierarchically using structural Verilog at the upper levels and behavioral Verilog for the leaf cells. For example, the controller module shows how a finite state machine is specified in behavioral Verilog and the aludec module shows how complex combinational logic is specified. The datapath is specified structurally in terms of wordslices, which are in turn described behaviorally.

For the sake of illustration, the 8-bit adder wordslice could be described structurally as a ripple carry adder composed of eight cascaded full adders. The full adder could be expressed structurally as a sum and a carry subcircuit. In turn, the sum and carry subcircuits could be expressed behaviorally. The full adder block is shown in Figure 1.59 while the carry subcircuit is explored further in Section 1.9.

```
module adder(input  logic [7:0] a, b,
             input  logic       c,
             output logic [7:0] s,
             output logic       cout);

  wire [6:0] carry;
```

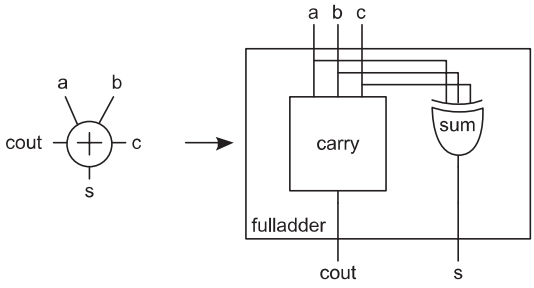


FIGURE 1.59 Full adder

```
  fulladder fa0(a[0], b[0], c,        s[0], carry[0]);
  fulladder fa1(a[1], b[1], carry[0], s[1], carry[1]);
  fulladder fa2(a[2], b[2], carry[1], s[2], carry[2]);
  ...
  fulladder fa7(a[7], b[7], carry[6], s[7], cout);
endmodule

module fulladder(input  logic a, b, c,
                 output logic s, cout);

  sum s1(a, b, c, s);
  carry c1(a, b, c, cout);
endmodule

module carry(input  logic a, b, c,
             output logic cout);

  assign cout = (a&b) | (a&c) | (b&c);
endmodule
```

1.9 Circuit Design

Circuit design is concerned with arranging transistors to perform a particular logic function. Given a circuit design, we can estimate the delay and power. The circuit can be represented as a schematic, or in textual form as a netlist. Common transistor level netlist formats include Verilog and SPICE. Verilog netlists are used for functional verification, while SPICE netlists have more detail necessary for delay and power simulations.

Because a transistor gate is a good insulator, it can be modeled as a capacitor, C. When the transistor is ON, some current I flows between source and drain. Both the current and capacitance are proportional to the transistor width.

The delay of a logic gate is determined by the current that it can deliver and the capacitance that it is driving, as shown in Figure 1.60 for one inverter driving another inverter. The capacitance is charged or discharged according to the constitutive equation

$$I = C\frac{dV}{dt}$$

If an average current I is applied, the time t to switch between 0 and V_{DD} is

$$t = \frac{C}{I}V_{DD}$$

Hence, the delay increases with the load capacitance and decreases with the drive current. To make these calculations, we will have to delve below the switch-level model of a transistor. Chapter 2 develops more detailed models of transistors accounting for the current and capacitance. One of the goals of circuit design is to choose transistor widths to meet delay requirements. Methods for doing so are discussed in Chapter 4.

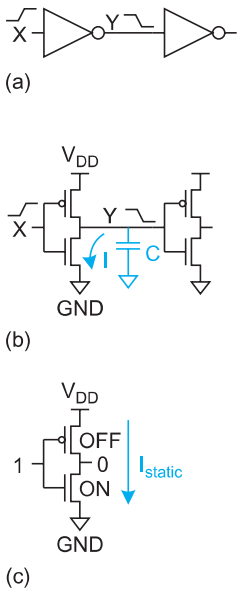


FIGURE 1.60 Circuit delay and power: (a) inverter pair, (b) transistor-level model showing capacitance and current during switching, (c) static leakage current during quiescent operation

Energy is required to charge and discharge the load capacitance. This is called dynamic power because it is consumed when the circuit is actively switching. The dynamic power consumed when a capacitor is charged and discharged at a frequency f is

$$P_{\text{dynamic}} = CV_{DD}^2 f$$

Even when the gate is not switching, it draws some static power. Because an OFF transistor is leaky, a small amount of current I_{static} flows between power and ground, resulting in a static power dissipation of

$$P_{\text{static}} = I_{\text{static}} V_{DD}$$

Chapter 5 examines power in more detail.

A particular logic function can be implemented in many ways. Should the function be built with ANDs, ORs, NANDs, or NORs? What should be the fan-in and fan-out of each gate? How wide should the transistors be on each gate? Each of these choices influences the capacitance and current and hence the speed and power of the circuit, as well as the area and cost.

As mentioned earlier, in many design methodologies, logic synthesis tools automatically make these choices, searching through the standard cells for the best implementation. For many applications, synthesis is good enough. When a system has critical requirements of high speed or low power or will be manufactured in large enough volume to justify the extra engineering, custom circuit design becomes important for critical portions of the chip.

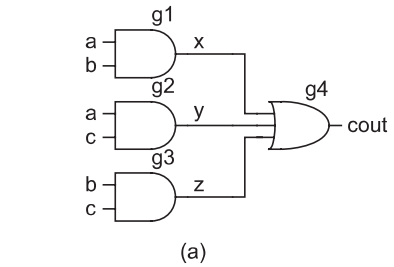
Circuit designers often draw schematics at the transistor and/or gate level. For example, Figure 1.61 shows two alternative circuit designs for the carry circuit in a full adder. The gate-level design in Figure 1.61(a) requires 26 transistors and four stages of gate delays (recall that ANDs and ORs are built from NANDs and NORs followed by inverters). The transistor-level design in Figure 1.61(b) requires only 12 transistors and two stages of gate delays, illustrating the benefits of optimizing circuit designs to take advantage of CMOS technology.

These schematics are then *netlisted* for simulation and verification. One common netlist format is structural Verilog HDL. The gate-level design can be netlisted as follows:

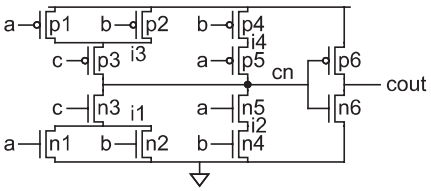
```
module carry(input  logic a, b, c,
             output logic cout);

  logic x, y, z;

  and g1(x, a, b);
  and g2(y, a, c);
  and g3(z, b, c);
  or  g4(cout, x, y, z);
endmodule
```



(a)



(b)

FIGURE 1.61 Carry subcircuit

This is a technology-independent structural description, because generic gates have been used and the actual gate implementations have not been specified. The transistor-level netlist follows:

```
module carry(input  logic a, b, c,
             output tri   cout);

  tri     i1, i2, i3, i4, cn;
  supply0 gnd;
  supply1 vdd;

  tranif1 n1(i1, gnd, a);
  tranif1 n2(i1, gnd, b);
  tranif1 n3(cn, i1, c);
  tranif1 n4(i2, gnd, b);
  tranif1 n5(cn, i2, a);
  tranif0 p1(i3, vdd, a);
  tranif0 p2(i3, vdd, b);
  tranif0 p3(cn, i3, c);
  tranif0 p4(i4, vdd, b);
  tranif0 p5(cn, i4, a);
  tranif1 n6(cout, gnd, cn);
  tranif0 p6(cout, vdd, cn);
endmodule
```

Transistors are expressed as

```
Transistor-type name(drain, source, gate);
```

`tranif1` corresponds to nMOS transistors that turn ON when the gate is 1 while `tranif0` corresponds to pMOS transistors that turn ON when the gate is 0. Appendix A.11 covers Verilog netlists in more detail.

With the description generated so far, we still do not have the information required to determine the speed or power consumption of the gate. We need to specify the size of the transistors and the stray capacitance. Because Verilog was designed as a switch-level and gate-level language, it is poorly suited to structural descriptions at this level of detail. Hence, we turn to another common structural language used by the circuit simulator SPICE. The specification of the transistor-level carry subcircuit at the circuit level might be represented as follows:

```
.SUBCKT CARRY A B C COUT VDD GND
MN1 I1 A GND GND NMOS W=2U L=0.6U AD=1.8P AS=3P
MN2 I1 B GND GND NMOS W=2U L=0.6U AD=1.8P AS=3P
MN3 CN C I1 GND NMOS W=2U L=0.6U AD=3P AS=3P
MN4 I2 B GND GND NMOS W=2U L=0.6U AD=0.9P AS=3P
MN5 CN A I2 GND NMOS W=2U L=0.6U AD=3P AS=0.9P
MP1 I3 A VDD VDD PMOS W=4U L=0.6U AD=3.6P AS=6P
MP2 I3 B VDD VDD PMOS W=4U L=0.6U AD=3.6P AS=6P
MP3 CN C I3 VDD PMOS W=4U L=0.6U AD=6P AS=6P
```

```
MP4 I4 B VDD VDD PMOS W=4U L=0.6U AD=1.8P AS=6P
MP5 CN A I4 VDD PMOS W=4U L=0.6U AD=6P AS=1.8P
MN6 COUT CN GND GND NMOS W=4U L=0.6U AD=6P AS=6P
MP6 COUT CN VDD VDD PMOS W=8U L=0.6U AD=12P AS=12P
CI1 I1 GND 6FF
CI3 I3 GND 9FF
CA A GND 12FF
CB B GND 12FF
CC C GND 6FF
CCN CN GND 12FF
CCOUT COUT GND 6FF
.ENDS
```

Transistors are specified by lines beginning with an M as follows:

```
Mname  drain   gate   source   body   type   W=width  L=length
       AD=drain area   AS=source area
```

Although MOS switches have been masquerading as three terminal devices (gate, source, and drain) until this point, they are in fact four terminal devices with the substrate or well forming the *body* terminal. The body connection was not listed in Verilog but is required for SPICE. The type specifies whether the transistor is a p-device or n-device. The width, length, and area parameters specify physical dimensions of the actual transistors. Units include U (micro, 10^{-6}), P (pico, 10^{-12}), and F (femto, 10^{-15}). Capacitors are specified by lines beginning with C as follows:

```
Cname   node1   node2   value
```

In this description, the MOS model in SPICE calculates the parasitic capacitances inherent in the MOS transistor using the device dimensions specified. The extra capacitance statements in the above description designate additional routing capacitance not inherent to the device structure. This depends on the physical design of the gate. Long wires also contribute resistance, which increases delay. At the circuit level of structural specification, all connections are given that are necessary to fully characterize the carry gate in terms of speed, power, and connectivity. Chapter 8 describes SPICE models in more detail.

1.10 Physical Design

1.10.1 Floorplanning

Physical design begins with a floorplan. The floorplan estimates the area of major units in the chip and defines their relative placements. The floorplan is essential to determine whether a proposed design will fit in the chip area budgeted and to estimate wiring lengths and wiring congestion. An initial floorplan should be prepared as soon as the logic is loosely defined. As usual, this process involves feedback. The floorplan will often suggest changes to the logic (and microarchitecture), which in turn changes the floorplan. For example, suppose microarchitects assume that a cache requires a 2-cycle access latency. If the floorplan shows that the data cache can be placed adjacent to the execution units in the

datapath, the cache access time might reduce to a single cycle. This could allow the microarchitects to reduce the cache capacity while providing the same performance. Once the cache shrinks, the floorplan must be reconsidered to take advantage of the newly available space near the datapath. As a complex design begins to stabilize, the floorplan is often hierarchically subdivided to describe the functional blocks within the units.

The challenge of floorplanning is estimating the size of each unit without proceeding through a detailed design of the chip (which would depend on the floorplan and wire lengths). This section assumes that good estimates have been made and describes what a floorplan looks like. The next sections describe each of the types of components that might be in a floorplan and suggests ways to estimate the component sizes.

Figure 1.62 shows the chip floorplan for the MIPS processor including the pad frame. The top-level blocks are the controller and datapath. A wiring channel is located between the two blocks to provide room to route control signals to the datapath. The datapath is further partitioned into wordslices. The *pad frame* includes 40 I/O pads, which are wired to the pins on the chip package. There are 29 pads used for signals; the remainder are V_{DD} and GND.

The floorplan is drawn to scale and annotated with dimensions. The chip is designed in a 0.6 μm process on a 1.5 × 1.5 mm die so the die is 5000 λ on a side. Each pad is 750 λ ×

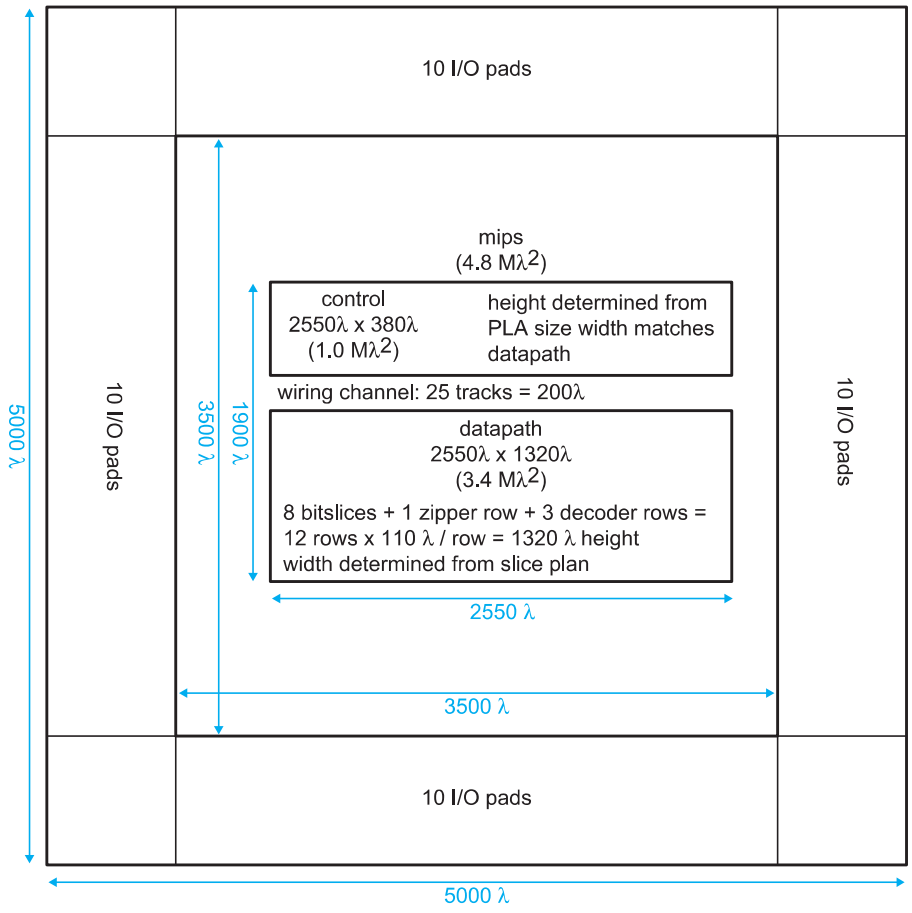


FIGURE 1.62 MIPS floorplan

350 λ, so the maximum possible core area inside the pad frame is 3500 $\lambda \times 3500\ \lambda = 12.25$ Mλ^2. Due to the wiring channel, the actual core area of 4.8 Mλ^2 is larger than the sum of the block areas. This design is said to be *pad-limited* because the I/O pads set the chip area. Most commercial chips are *core-limited* because the chip area is set by the logic excluding the pads. In general, blocks in a floorplan should be rectangular because it is difficult for a designer to stuff logic into an odd-shaped region (although some CAD tools do so just fine).

Figure 1.63 shows the actual chip layout. Notice the 40 I/O pads around the periphery. Just inside the pad frame are metal2 V_{DD} and GND rings, marked with + and –.

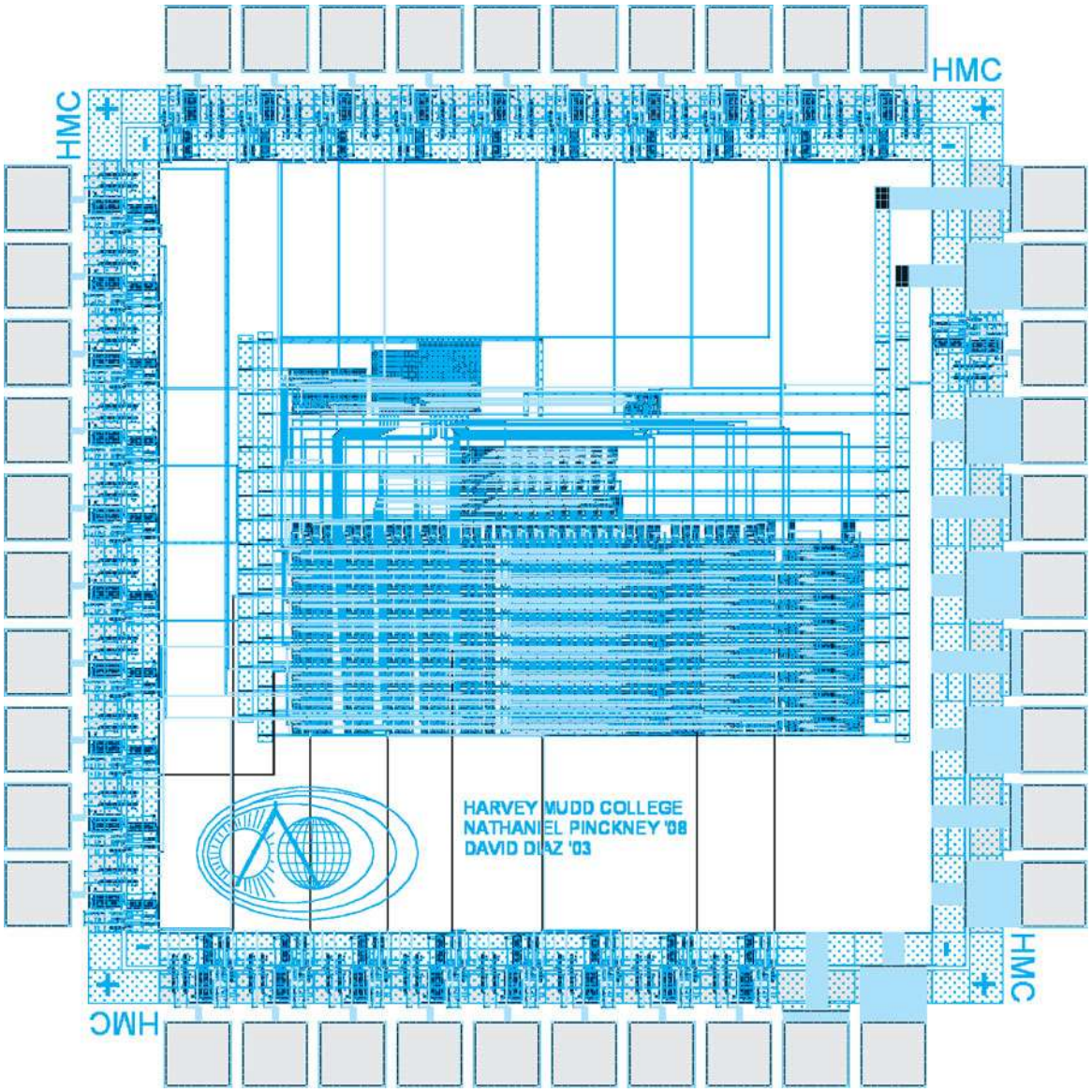


FIGURE 1.63 MIPS layout

On-chip structures can be categorized as *random logic*, *datapaths*, *arrays*, *analog*, and *input/output* (I/O). Random logic, like the aludecoder, has little structure. Datapaths operate on multi-bit data words and perform roughly the same function on each bit so they consist of multiple *N*-bit wordslices. Arrays, like RAMs, ROMs, and PLAs, consist of identical cells repeated in two dimensions. Productivity is highest if layout can be reused or automatically generated. Datapaths and arrays are good VLSI building blocks because a single carefully crafted cell is reused in one or two dimensions. Automatic layout generators exist for memory arrays and random logic but are not as mature for datapaths. Therefore, many design methodologies ignore the potential structure of datapaths and instead lay them out with random logic tools except when performance or area are vital. Analog circuits still require careful design and simulation but tend to involve only small amounts of layout because they have relatively few transistors. I/O cells are also highly tuned to each fabrication process and are often supplied by the process vendor.

Random logic and datapaths are typically built from *standard cells* such as inverters, NAND gates, and flip-flops. Standard cells increase productivity because each cell only needs to be drawn and verified once. Often, a standard cell library is purchased from a third party vendor.

Another important decision during floorplanning is to choose the metal orientation. The MIPS floorplan uses horizontal metal1 wires, vertical metal2 wires, and horizontal metal3 wires. Alternating directions between each layer makes it easy to cross wires on different layers.

1.10.2 Standard Cells

A simple standard cell library is shown on the inside front cover. Power and ground run horizontally in metal1. These supply rails are 8 λ wide (to carry more current) and are separated by 90 λ center-to-center. The nMOS transistors are placed in the bottom 40 λ of the cell and the pMOS transistors are placed in the top 50 λ. Thus, cells can be connected by abutment with the supply rails and n-well matching up. Substrate and well contacts are placed under the supply rails. Inputs and outputs are provided in metal2, which runs vertically. Each cell is a multiple of 8 λ in width so that it offers an integer number of metal2 tracks. Within the cell, poly is run vertically to form gates and diffusion and metal1 are run horizontally, though metal1 can also be run vertically to save space when it does not interfere with other connections.

Cells are tiled in rows. Each row is separated vertically by at least 110 λ from the base of the previous row. In a 2-level metal process, horizontal metal1 wires are placed in *routing channels* between the rows. The number of wires that must be routed sets the height of the routing channels. Layout is often generated with automatic place & route tools. Figure 1.64 shows the controller layout generated by such a tool. Note that in this and subsequent layouts, the n-well around the pMOS transistors will usually not be shown.

When more layers of metal are available, routing takes place over the cells and routing channels may become unnecessary. For example, in a 3-level metal process, metal3 is run horizontally on a 10 λ pitch. Thus, 11 horizontal tracks can run over each cell. If this is sufficient to accommodate all of the horizontal wires, the routing channels can be eliminated.

Automatic synthesis and place & route tools have become good enough to map entire designs onto standard cells. Figure 1.65 shows the entire 8-bit MIPS processor synthesized from the VHDL model given in Appendix A.12 onto a cell library in a 130 nm process with

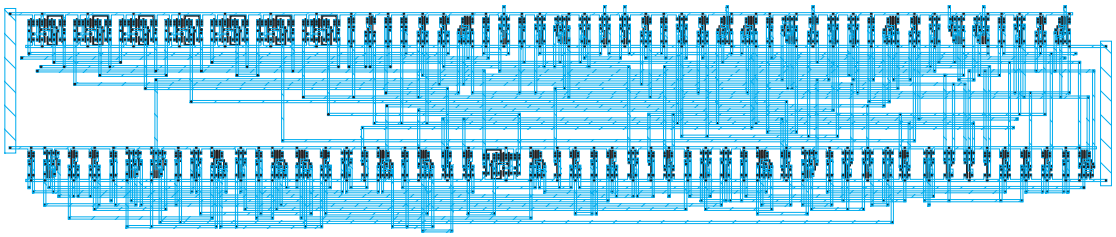


FIGURE 1.64 MIPS controller layout (synthesized)

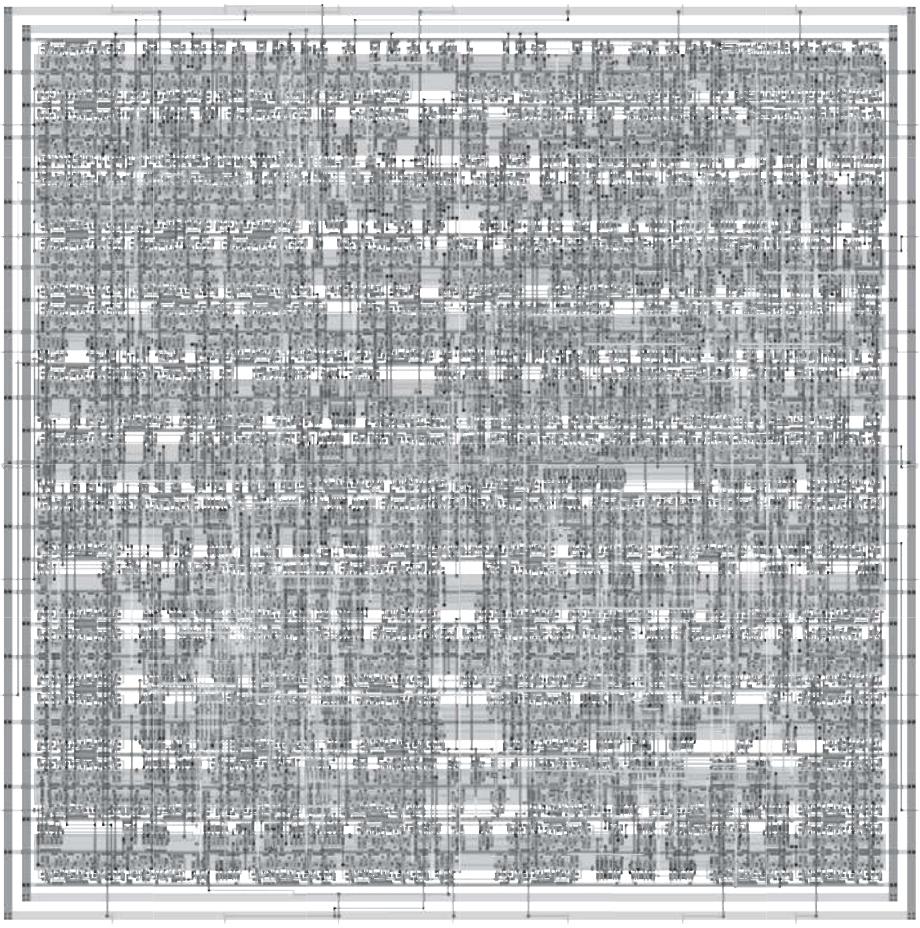


FIGURE 1.65 Synthesized MIPS processor

seven metal layers. Compared to Figure 1.63, the synthesized design shows little discernible structure except that 26 rows of standard cells can be identified beneath the wires. The area is approximately 4 Mλ^2. Synthesized designs tend to be somewhat slower than a good custom design, but they also take an order of magnitude less design effort.

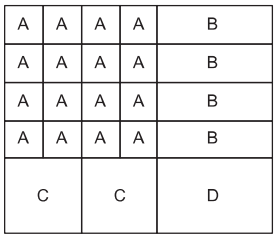


FIGURE 1.66 Pitch-matching of snap-together cells

1.10.3 Pitch Matching

The area of the controller in Figure 1.64 is dominated by the routing channels. When the logic is more regular, layout density can be improved by including the wires in cells that "snap together." Snap-together cells require more design and layout effort but lead to smaller area and shorter (i.e., faster) wires. The key issue in designing snap-together cells is *pitch-matching*. Cells that connect must have the same size along the connecting edge. Figure 1.66 shows several pitch-matched cells. Reducing the size of cell *D* does not help the layout area. On the other hand, increasing the size of cell *D* also affects the area of *B* and/or *C*.

Figure 1.67 shows the MIPS datapath in more detail. The eight horizontal bitslices are clearly visible. The zipper at the top of the layout includes three rows for the decoder that is pitch-matched to the register file in the datapath. Vertical metal2 wires are used for control, including clocks, multiplexer selects, and register enables. Horizontal metal3 wires run over the tops of cells to carry data along a bitslice.

The width of the transistors in the cells and the number of wires that must run over the datapath determines the minimum height of the datapath cells. 60–100 λ are typical heights for relatively simple datapaths. The width of the cell depends on the cell contents.

1.10.4 Slice Plans

Figure 1.68 shows a *slice plan* of the datapath. The diagram illustrates the ordering of wordslices and the allocation of wiring tracks within each bitslice. Dots indicate that a bus passes over a cell and is also used in that cell. Each cell is annotated with its type and width (in number of tracks). For example, the program counter (`pc`) is an output of the PC flop and is also used as an input to the srcA and address multiplexers. The slice plan

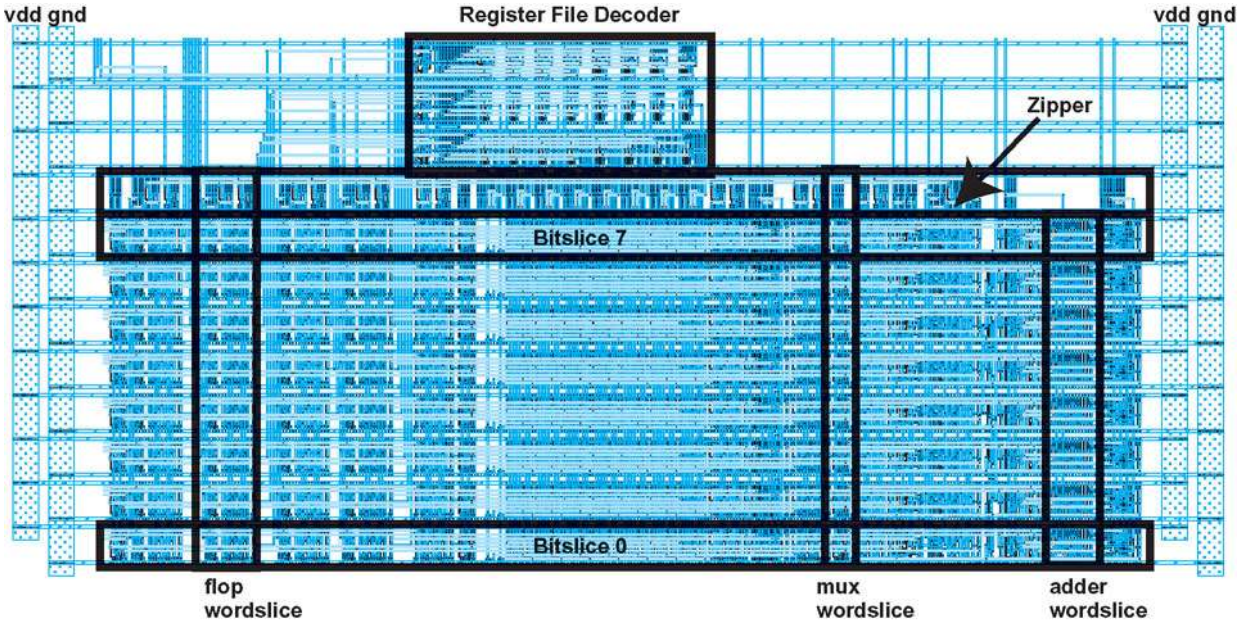


FIGURE 1.67 MIPS datapath layout

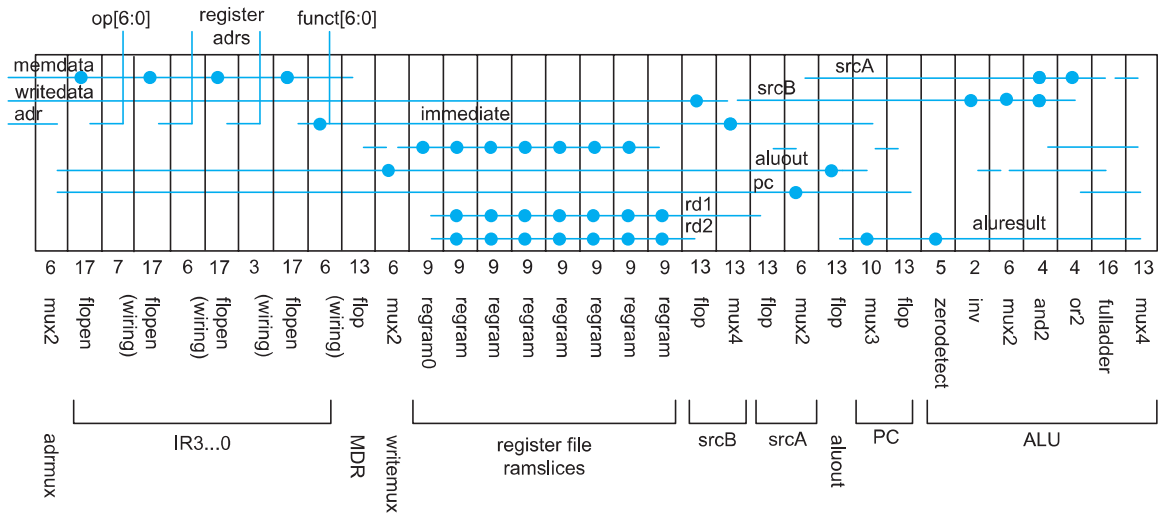


FIGURE 1.68 Datapath slice plan

makes it easy to calculate wire lengths and evaluate wiring congestion before laying out the datapath. In this case, it is evident that the greatest congestion takes place over the register file, where seven wiring tracks are required.

The slice plan is also critical for estimating area of datapaths. Each wordslice is annotated with its width, measured in tracks. This information can be obtained by looking at the cell library layouts. By adding up the widths of each element in the slice plan, we see that the datapath is 319 tracks wide, or 2552 λ wide. There are eight bitslices in the 8-bit datapath. In addition, there is one more row for the zipper and three more for the three register file address decoders, giving a total of 12 rows. At a pitch of 110 λ / row, the datapath is 1320 λ tall. The address decoders only occupy a small fraction of their rows, leaving wasted empty space. In a denser design, the controller could share some of the unused area.

1.10.5 Arrays

Figure 1.69 shows a programmable logic array (PLA) used for the control FSM next state and output logic. A PLA can compute any function expressed in sum of products form. The structure on the left is called the AND plane and the structure on the right is the OR plane. PLAs are discussed further in Section 12.7.

This PLA layout uses 2 vertical tracks for each input and 3 for each output plus about 6 for overhead. It uses 1.5 horizontal tracks for each product or minterm, plus about 14 for overhead. Hence, the size of a PLA is easy to calculate. The total PLA area is 500 $\lambda \times 350$ λ, plus another 336 $\lambda \times 220$ λ for the four external flip-flops needed in the control FSM. The height of the controller is dictated by the height of the PLA plus a few wiring tracks to route inputs and outputs. In comparison, the synthesized controller from Figure 1.64 has a size of 1500 $\lambda \times 400$ λ because the wiring tracks waste so much space.

1.10.6 Area Estimation

A good floorplan depends on reasonable area estimates, which may be difficult to make before logic is finalized. An experienced designer may be able to estimate block area by

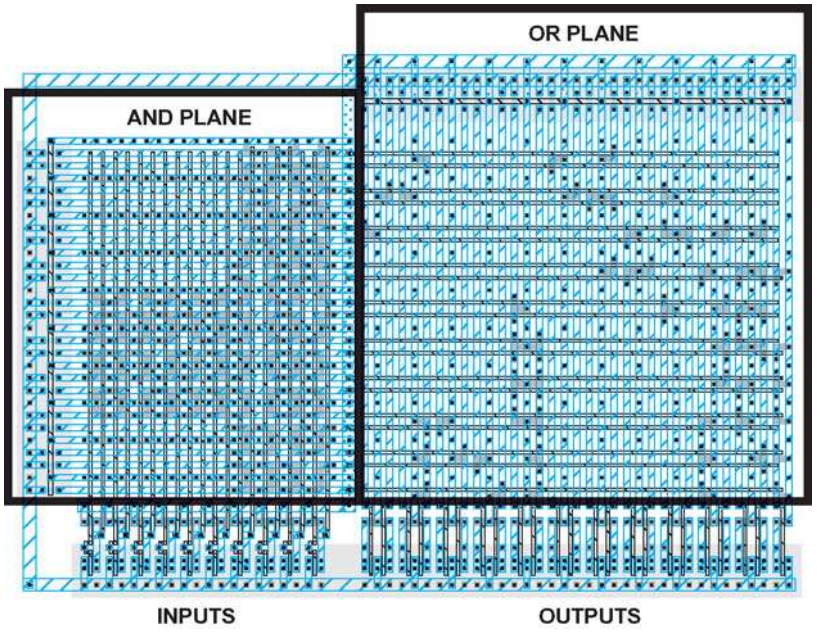


FIGURE 1.69 PLA for control FSM

comparison to the area of a comparable block drawn in the past. In the absence of data for such comparison, Table 1.10 lists some typical numbers. Be certain to account for large wiring channels at a pitch of 8 λ / track. Larger transistors clearly occupy a greater area, so this may be factored into the area estimates as a function of W and L (width and length). For memories, don't forget about the decoders and other periphery circuits, which often take as much area as the memory bits themselves. Your mileage may vary, but datapaths and arrays typically achieve higher densities than standard cells.

TABLE 1.10 Typical layout densities

Element	Area
random logic (2-level metal process)	1000 – 1500 λ^2 / transistor
datapath	250 – 750 λ^2 / transistor or 6 WL + 360 λ^2 / transistor
SRAM	1000 λ^2 / bit
DRAM (in a DRAM process)	100 λ^2 / bit
ROM	100 λ^2 / bit

Given enough time, it is nearly always possible to shave a few lambda here or there from a design. However, such efforts are seldom a good investment unless an element is repeated so often that it accounts for a major fraction of the chip area or if floorplan errors have led to too little space for a block and the block must be shrunk before the chip can be completed. It is wise to make conservative area estimates in floorplans, especially if there is risk that more functionality may be added to a block.

Some cell library vendors specify typical routed standard cell layout densities in kgates / mm^2.[7] Commonly, a gate is defined as a 3-input static CMOS NAND or NOR with six transistors. A 65 nm process ($\lambda \approx 0.03\ \mu$m) with eight metal layers may achieve a density of 160–500 kgates / mm^2 for random logic. This corresponds to about 370–1160 λ^2 / transistor. Processes with many metal layers obtain high density because routing channels are not needed.

1.11 Design Verification

Integrated circuits are complicated enough that if anything can go wrong, it probably will. Design verification is essential to catching the errors before manufacturing and commonly accounts for half or more of the effort devoted to a chip.

As design representations become more detailed, verification time increases. It is not practical to simulate an entire chip in a circuit-level simulator such as SPICE for a large number of cycles to prove that the layout is correct. Instead, the design is usually tested for functionality at the architectural level with a model in a language such as C and at the logic level by simulating the HDL description. Then, the circuits are checked to ensure that they are a faithful representation of the logic and the layout is checked to ensure it is a faithful representation of the circuits, as shown in Figure 1.70. Circuits and layout must meet timing and power specifications as well.

A *testbench* is used to verify that the logic is correct. The testbench instantiates the logic under test. It reads a file of inputs and expected outputs called *test vectors*, applies them to the module under test, and logs mismatches. Appendix A.12 provides an example of a testbench for verifying the MIPS processor logic.

A number of techniques are available for circuit verification. If the logic is synthesized onto a cell library, the postsynthesis gate-level netlist can be expressed in an HDL again and simulated using the same test vectors. Alternatively, a transistor-level netlist can be simulated against the test vector, although this can result in tricky race conditions for sequential circuits. Powerful *formal verification* tools are also available to check that a circuit performs the same Boolean function as the associated logic. Exotic circuits should be simulated thoroughly to ensure that they perform the intended logic function and have adequate noise margins; circuit pitfalls are discussed throughout this book.

Layout vs. Schematic tools (LVS) check that transistors in a layout are connected in the same way as in the circuit schematic. *Design rule checkers* (DRC) verify that the layout satisfies design rules. *Electrical rule checkers* (ERC) scan for other potential problems such as noise or premature wearout; such problems will also be discussed later in the book.

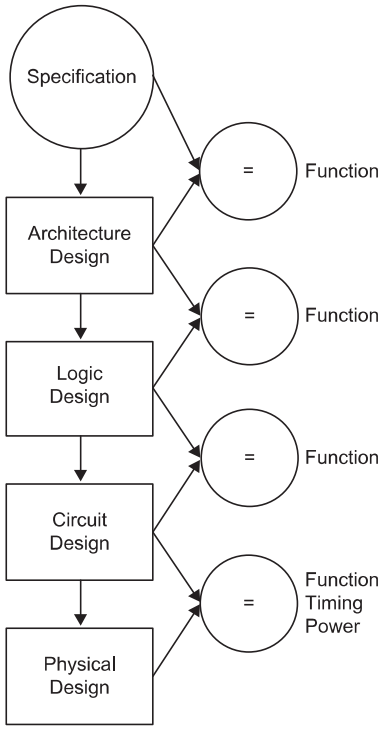


FIGURE 1.70 Design and verification sequence

[7] 1 kgate = 1000 gates.

1.12 Fabrication, Packaging, and Testing

Once a chip design is complete, it is taped out for manufacturing. *Tapeout* gets its name from the old practice of writing a specification of masks to magnetic tape; today, the mask descriptions are usually sent to the manufacturer electronically. Two common formats for mask descriptions are the Caltech Interchange Format (CIF) [Mead80] (mainly used in academia) and the Calma GDS II Stream Format (GDS) [Calma84] (used in industry).

Masks are made by etching a pattern of chrome on glass with an electron beam. A set of masks for a nanometer process can be very expensive. For example, masks for a large chip in a 180 nm process may cost on the order of a quarter of a million dollars. In a 65 nm process, the mask set costs about $3 million. The MOSIS service in the United States and its EUROPRACTICE and VDEC counterparts in Europe and Japan make a single set of masks covering multiple small designs from academia and industry to amortize the cost across many customers. With a university discount, the cost for a run of 40 small chips on a multi-project wafer can run about $10,000 in a 130 nm process down to $2000 in a 0.6 μm process. MOSIS offers certain grants to cover fabrication of class project chips.

Integrated circuit fabrication plants (fabs) now cost billions of dollars and become obsolete in a few years. Some large companies still own their own fabs, but an increasing number of fabless semiconductor companies contract out manufacturing to foundries such as TSMC, UMC, and IBM.

Multiple chips are manufactured simultaneously on a single silicon wafer, typically 150–300 mm (6″–12″) in diameter. Fabrication requires many deposition, masking, etching, and implant steps. Most fabrication plants are optimized for wafer throughput rather than latency, leading to turnaround times of up to 10 weeks. Figure 1.71 shows an engineer in a *clean room* holding a completed 300 mm wafer. Clean rooms are filtered to eliminate most dust and other particles that could damage a partially processed wafer. The engineer is wearing a "bunny suit" to avoid contaminating the clean room. Figure 1.72 is a

FIGURE 1.71 Engineer holding processed 12-inch wafer (Photograph courtesy of the Intel Corporation.)

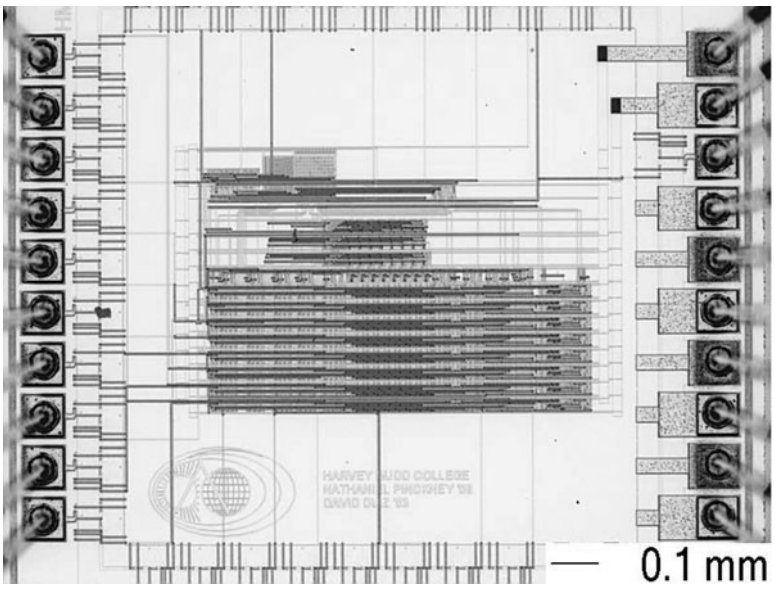


FIGURE 1.72 MIPS processor photomicrograph (only part of pad frame shown)

photomicrograph (a photograph taken under a microscope) of the 8-bit MIPS processor.

Processed wafers are sliced into dice (chips) and packaged. Figure 1.73 shows the 1.5 × 1.5 mm chip in a 40-pin *dual-inline package* (DIP). This *wire-bonded* package uses thin gold wires to connect the pads on the die to the lead frame in the center cavity of the package. These wires are visible on the pads in Figure 1.72. More advanced packages offer different trade-offs between cost, pin count, pin bandwidth, power handling, and reliability, as will be discussed in Section 13.2. Flip-chip technology places small solder balls directly onto the die, eliminating the bond wire inductance and allowing contacts over the entire chip area rather than just at the periphery.

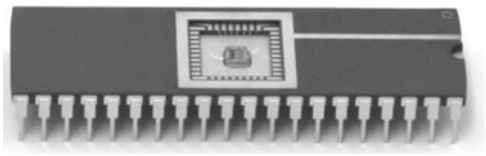


FIGURE 1.73 Chip in a 40-pin dual-inline package

Even tiny defects in a wafer or dust particles can cause a chip to fail. Chips are tested before being sold. Testers capable of handling high-speed chips cost millions of dollars, so many chips use built-in self-test features to reduce the tester time required. Chapter 15 is devoted to design verification and testing.

Summary and a Look Ahead

> *"If the automobile had followed the same development cycle as the computer, a Rolls-Royce would today cost $100, get one million miles to the gallon, and explode once a year . . ."*
>
> —Robert X. Cringely

CMOS technology, driven by Moore's Law, has come to dominate the semiconductor industry. This chapter examined the principles of designing a simple CMOS integrated circuit. MOS transistors can be viewed as electrically controlled switches. Static CMOS gates are built from pull-down networks of nMOS transistors and pull-up networks of pMOS transistors. Transistors and wires are fabricated on silicon wafers using a series of deposition, lithography, and etch steps. These steps are defined by a set of masks drawn as a chip layout. Design rules specify minimum width and spacing between elements in the layout. The chip design process can be divided into architecture, logic, circuit, and physical design. The performance, area, and power of the chip are influenced by interrelated decisions made at each level. Design verification plays an important role in constructing such complex systems; the reliability requirements for hardware are much greater than those typically imposed on software.

Primary design objectives include reliability, performance, power, and cost. Any chip should, with high probability, operate reliably for its intended lifetime. For example, the chip must be designed so that it does not overheat or break down from excessive voltage. Performance is influenced by many factors including clock speed and parallelism. CMOS transistors dissipate power every time they switch, so the dynamic power consumption is related to the number and size of transistors and the rate at which they switch. At feature

sizes below 180 nm, transistors also leak a significant amount of current even when they should be OFF. Thus, chips now draw static power even when they are idle. One of the central challenges of VLSI design is making good trade-offs between performance and power for a particular application. The cost of a chip includes nonrecurring engineering (NRE) expenses for the design and masks, along with per-chip manufacturing costs related to the size of the chip. In processes with smaller feature sizes, the per-unit cost goes down because more transistors can be packed into a given area, but the NRE increases. The latest manufacturing processes are only cost-effective for chips that will sell in huge volumes. Nevertheless, plenty of interesting markets exist for chips in mature, inexpensive manufacturing processes.

To quantify how a chip meets these objectives, we must develop and analyze more complete models. The remainder of this book will expand on the material introduced in this chapter. Of course, transistors are not simply switches. Chapter 2 examines the current and capacitance of transistors, which are essential for estimating delay and power. A more detailed description of CMOS processing technology and layout rules is presented in Chapter 3. The next four chapters address the fundamental concerns of circuit designers. The models from Chapter 2 are too detailed to apply by hand to large systems, yet not detailed enough to fully capture the complexity of modern transistors. Chapter 4 develops simplified models to estimate the delay of circuits. If modern chips were designed to squeeze out the ultimate possible performance without regard to power, they would burn up. Thus, it is essential to estimate and trade off the power consumption against performance. Moreover, low power consumption is crucial to mobile battery-operated systems. Power is considered in Chapter 5. Wires are as important as transistors in their contribution to overall performance and power, and are discussed in Chapter 6. Chapter 7 addresses design of robust circuits with a high yield and low failure rate.

Simulation is discussed in Chapter 8 and is used to obtain more accurate performance and power predictions as well as to verify the correctness of circuits and logic. Chapter 9 considers combinational circuit design. A whole kit of circuit families are available with different trade-offs in speed, power, complexity, and robustness. Chapter 10 continues with sequential circuit design, including clocking and latching techniques.

The next three chapters delve into CMOS subsystems. Chapter 11 catalogs designs for a host of datapath subsystems including adders, shifters, multipliers, and counters. Chapter 12 similarly describes memory subsystems including SRAMs, DRAMs, CAMs, ROMs, and PLAs. Chapter 13 addresses special-purpose subsystems including power distribution, clocking, and I/O.

The final chapters address practicalities of CMOS system design. Chapter 14 focuses on a range of current design methods, identifying the issues peculiar to CMOS. Testing, design-for-test, and debugging techniques are discussed in Chapter 15. Hardware description languages (HDLs) are used in the design of nearly all digital integrated circuits today. Appendix A provides side-by-side tutorials for Verilog and VHDL, the two dominant HDLs.

A number of sections are marked with an "optional" icon. These sections describe particular subjects in greater detail. You may skip over these sections on a first reading and return to them when they are of practical relevance. To keep the length of this book under control, some optional topics have been published on the Internet rather than in print. These sections can be found at `www.cmosvlsi.com` and are labeled with a "Web Enhanced" icon. A companion text, *Digital VLSI Chip Design with Cadence and Synopsys CAD Tools* [Brunvand09], covers practical details of using the leading industrial CAD tools to build chips.

Exercises

1.1 Extrapolating the data from Figure 1.4, predict the transistor count of a microprocessor in 2016.

1.2 Search the Web for transistor counts of Intel's more recent microprocessors. Make a graph of transistor count vs. year of introduction from the Pentium Processor in 1993 to the present on a semilogarithmic scale. How many months pass between doubling of transistor counts?

1.3 As the cost of a transistor drops from a microbuck ($\$10^{-6}$) toward a nanobuck, what opportunities can you imagine to change the world with integrated circuits?

1.4 Read a biography or history about a major event in the development of integrated circuits. For example, see *Crystal Fire* by Lillian Hoddesonor, *Microchip* by Jeffrey Zygmont, or *The Pentium Chronicles* by Robert Colwell. Pick a team or individual that made a major contribution to the field. In your opinion, what were the characteristics that led to success? What traits of the team management would you seek to emulate or avoid in your own professional life?

1.5 Sketch a transistor-level schematic for a CMOS 4-input NOR gate.

1.6 Sketch a transistor-level schematic for a compound CMOS logic gate for each of the following functions:

a) $Y = \overline{ABC + D}$

b) $Y = \overline{(AB + C) \cdot D}$

c) $Y = \overline{AB + C \cdot (A + B)}$

1.7 Use a combination of CMOS gates (represented by their symbols) to generate the following functions from A, B, and C.

a) $Y = A$ (buffer)

b) $Y = A\overline{B} + \overline{A}B$ (XOR)

c) $Y = \overline{A}\overline{B} + AB$ (XNOR)

d) $Y = AB + BC + AC$ (majority)

1.8 Sketch a transistor-level schematic of a CMOS 3-input XOR gate. You may assume you have both true and complementary versions of the inputs available.

1.9 Sketch transistor-level schematics for the following logic functions. You may assume you have both true and complementary versions of the inputs available.

a) A 2:4 decoder defined by

$Y0 = \overline{A0} \cdot \overline{A1}$
$Y1 = A0 \cdot \overline{A1}$
$Y2 = \overline{A0} \cdot A1$
$Y3 = A0 \cdot A1$

b) A 3:2 priority encoder defined by

$Y0 = \overline{A0} \cdot (A1 + \overline{A2})$
$Y1 = \overline{A0} \cdot \overline{A1}$

1.10 Sketch a stick diagram for a CMOS 4-input NOR gate from Exercise 1.5.

1.11 Estimate the area of your 4-input NOR gate from Exercise 1.10.

1.12 Using a CAD tool of your choice, layout a 4-input NOR gate. How does its size compare to the prediction from Exercise 1.11?

1.13 Figure 1.74 shows a stick diagram of a 2-input NAND gate. Sketch a side view (cross-section) of the gate from X to X′.

1.14 Figure 1.75 gives a stick diagram for a level-sensitive latch. Estimate the area of the latch.

1.15 Draw a transistor-level schematic for the latch of Figure 1.75. How does the schematic differ from Figure 1.31(b)?

1.16 Consider the design of a CMOS compound OR-AND-INVERT (OAI21) gate computing $F = \overline{(A + B) \cdot C}$.

a) sketch a transistor-level schematic

b) sketch a stick diagram

c) estimate the area from the stick diagram

d) layout your gate with a CAD tool using unit-sized transistors

e) compare the layout size to the estimated area

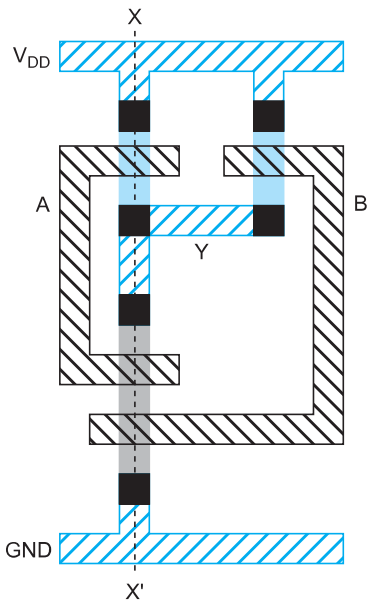


FIGURE 1.74 2-input NAND gate stick diagram

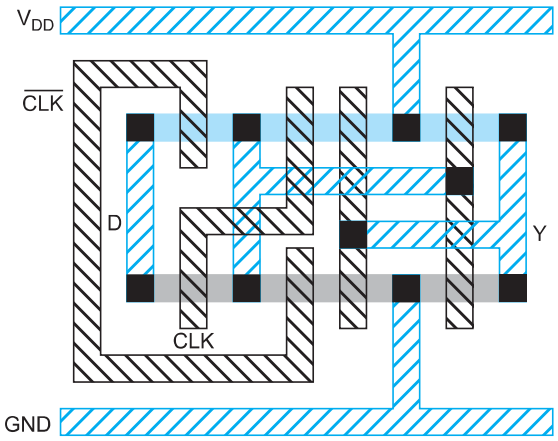


FIGURE 1.75 Level-sensitive latch stick diagram

1.17 Consider the design of a CMOS compound OR-OR-AND-INVERT (OAI22) gate computing $F = \overline{(A + B) \cdot (C + D)}$.

a) sketch a transistor-level schematic

b) sketch a stick diagram

c) estimate the area from the stick diagram

d) layout your gate with a CAD tool using unit-sized transistors

e) compare the layout size to the estimated area

1.18 A 3-input majority gate returns a true output if at least two of the inputs are true. A minority gate is its complement. Design a 3-input CMOS minority gate using a single stage of logic.

a) sketch a transistor-level schematic

b) sketch a stick diagram

c) estimate the area from the stick diagram

1.19 Design a 3-input minority gate using CMOS NANDs, NORs, and inverters. How many transistors are required? How does this compare to a design from Exercise 1.18(a)?

1.20 A carry lookahead adder computes $G = G_3 + P_3(G_2 + P_2(G_1 + P_1G_0))$. Consider designing a compound gate to compute $\overline{G}$.

a) sketch a transistor-level schematic

b) sketch a stick diagram

c) estimate the area from the stick diagram

1.21 `www.cmosvlsi.com` has a series of four labs in which you can learn VLSI design by completing the multicycle MIPS processor described in this chapter. The labs use the open-source Electric CAD tool or commercial tools from Cadence and Synopsys. They cover the following:

a) leaf cells: schematic entry, layout, icons, simulation, DRC, ERC, LVS; hierarchical design

b) datapath design: wordslices, ALU assembly, datapath routing

c) control design: random logic or PLAs

d) chip assembly, pad frame, global routing, full-chip verification, tapeout

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